

Probability Concepts and Random Variable

Random Experiment

An experiment whose outcome or result can be predicted with certainty is called a Deterministic experiment.

Although all possible outcomes of an experiment may be known in advance the outcome of a particular performance of the experiment cannot be predicted owing to a number of unknown causes. Such an experiment is called a Random experiment.

(e.g.) Whenever a fair dice is thrown, it is known that any of the 6 possible outcomes will occur, but it cannot be predicted what exactly the outcome will be.

Sample Space

The set of all possible outcomes which are assumed equally likely.

Event

A sub-set of S consisting of possible outcomes.

Mathematical definition of Probability

Let S be the sample space and A be an event associated with a random experiment. Let $n(S)$ and $n(A)$ be the number of elements of S and A. then the probability of event A occurring is denoted as $P(A)$, is denoted by

$$P(A) = \frac{n(A)}{n(S)}$$

Note: 1. It is obvious that $0 \leq P(A) \leq 1$.

2. If A is an impossible event, $P(A) = 0$.

3. If A is a certain event, $P(A) = 1$.

A set of events is said to be mutually exclusive if the occurrence of any one of them excludes the occurrence of the others. That is, set of the events does not occur simultaneously,

$P(A_1 \cap A_2 \cap A_3 \cap \dots \cap A_n, \dots) = 0$ A set of events is said to be mutually exclusive if the occurrence of any one of them excludes the occurrence of the others. That is, set of the events does not occur simultaneously,

$$P(A_1 \cap A_2 \cap A_3 \cap \dots \cap A_n, \dots) = 0$$

Axiomatic definition of Probability

Let S be the sample space and A be an event associated with a random experiment. Then the probability of the event A, $P(A)$ is defined as a real number satisfying the following axioms.

1. $0 \leq P(A) \leq 1$

2. $P(S) = 1$

3. If A and B are mutually exclusive events, $P(A \cup B) = P(A) + P(B)$ and

4. If $A_1, A_2, A_3, \dots, A_n, \dots$ are mutually exclusive events,

$$P(A_1 \cup A_2 \cup A_3 \cup \dots \cup A_n, \dots) = P(A_1) + P(A_2) + P(A_3) + \dots + P(A_n), \dots$$

Important Theorems

Theorem 1: Probability of impossible event is zero.

Proof: Let S be sample space (certain events) and ϕ be the impossible event.

Certain events and impossible events are **mutually exclusive**.

$$P(S \cup \phi) = P(S) + P(\phi) \quad (\text{Axiom 3})$$

$$S \cup \phi = S$$

$$P(S) = P(S) + P(\phi)$$

$$P(\phi) = 0, \text{ hence the result.}$$

Theorem 2: If A is the complementary event of A , $P(A) = 1 - P(A) \leq 1$.

Proof: Let A be the occurrence of the event —

A be the non-occurrence of the event .

Occurrence and non-occurrence of the event are **mutually exclusive**.

$$P(A \cup A) = P(A) + P(A)$$

$$A \cup A = S \quad \Rightarrow \quad P(A \cup A) = P(S) = 1$$

$$\therefore 1 = P(A) + P(A) \quad —$$

$$P(A) = 1 - P(A) \leq 1. \quad —$$

Theorem-3: (Addition theorem)

If A and B are any 2 events,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \leq P(A) + P(B).$$

Proof: We know, $A = AB \cup AB$ and $B = AB \cup AB$

$$\therefore P(A) = P(\overline{AB}) + P(AB) \text{ and } P(B) = P(\overline{AB}) + P(AB) \quad (\text{Axiom 3})$$

$$P(A) + P(B) = P(\overline{AB}) + P(AB) + P(\overline{AB}) + P(AB)$$

$$= P(A \cup B) + P(A \cap B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \leq P(A) + P(B).$$

Note: The theorem can be extended to any 3 events, A, B and C

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

Theorem 4: If $B \subset A$, $P(B) \leq P(A)$.

Proof: A and $\overline{AB}$ are **mutually exclusive** events such that $B \cup \overline{AB} = A$

$$\therefore P(B \cup \overline{AB}) = P(A)$$

$$P(B) + P(\overline{AB}) = P(A) \quad (\text{Axiom 3})$$

$$P(B) \leq P(A)$$

Conditional Probability

The conditional probability of an event B , assuming that the event A has happened, is denoted by $P(B/A)$ and defined as

$$P(B/A) = \frac{P(A \cap B)}{P(A)}, \text{ provided } P(A) \neq 0$$

Product theorem of probability

Rewriting the definition of conditional probability, We get

$$P(A \cap B) = P(A)P(A/B)$$

The product theorem can be extended to 3 events, A, B and C as follows:

$$P(A \cap B \cap C) = P(A)P(B/A)P(C/A \cap B)$$

Note: 1. If $A \subset B$, $P(B/A) = 1$, since $A \cap B = A$.

2. If $B \subset A$, $P(B/A) \geq P(B)$, since $A \cap B = B$, and $\frac{P(B)}{P(A)} \geq P(B)$,

As $P(A) \leq P(S) = 1$.

3. If A and B are **mutually exclusive** events, $P(B/A) = 0$, since $P(A \cap B) = 0$.

4. If $P(A) > P(B)$, $P(A/B) > P(B/A)$.

5. If $A_1 \subset A_2$, $P(A_1/B) \leq P(A_2/B)$.

Independent Events

A set of events is said to be independent if the occurrence of any one of them does not depend on the occurrence or non-occurrence of the others.

If the two events A and B are independent, the product theorem takes the form $P(A \cap B) = P(A) \times P(B)$, Conversely, if $P(A \cap B) = P(A) \times P(B)$, the events are said to be independent (pair wise independent).

The product theorem can be extended to any number of independent events, If $A_1 A_2 A_3 \dots A_n$ are n independent events, then

$$P(A_1 \cap A_2 \cap A_3 \cap \dots \cap A_n) = P(A_1) \times P(A_2) \times P(A_3) \times \dots \times P(A_n)$$

Theorem 4:

If the events A and B are independent, the events $\bar{A}$ and B are also [independent](#).

Proof:

The events $A \cap B$ and $\bar{A} \cap B$ are mutually exclusive such that $(A \cap B) \cup (\bar{A} \cap B) = B$

$$\begin{aligned} \therefore P(A \cap B) + P(\bar{A} \cap B) &= P(B) \\ P(\bar{A} \cap B) &= P(B) - P(A \cap B) \\ &= P(B) - P(A)P(B) \quad (\because A \text{ and } B \text{ are independent}) \\ &= P(B)[1 - P(A)] \\ &= P(\bar{A})P(B). \end{aligned}$$

Theorem 5:

If the events A and B are independent, the events $\bar{A}$ and $\bar{B}$ are also [independent](#).

Proof:

$$\begin{aligned} P(\bar{A} \cap \bar{B}) &= P(\overline{A \cup B}) = 1 - P(A \cup B) \\ &= 1 - [P(A) + P(B) - P(A \cap B)] \quad (\text{Addition theorem}) \\ &= [1 - P(A)] - P(B)[1 - P(A)] \\ &= P(\bar{A})P(\bar{B}). \end{aligned}$$

Problem 1:

From a bag containing 3 red and 2 black balls, 2 balls are drawn at random. Find the probability that they are of the same colour.

Solution :

Let A be the event of drawing 2 red balls

B be the event of drawing 2 black balls.

$$\begin{aligned}\therefore P(A \cup B) &= P(A) + P(B) \\ &= \frac{{}^3C_2}{{}^5C_2} + \frac{{}^2C_2}{{}^5C_2} = \frac{3}{10} + \frac{1}{10} = \frac{2}{5}\end{aligned}$$

Problem 2:

When 2 cards are drawn from a well-shuffled pack of playing cards, what is the probability that they are of the same suit?

Solution :

Let A be the event of drawing 2 spade cards

B be the event of drawing 2 club cards

C be the event of drawing 2 hearts cards

D be the event of drawing 2 diamond cards.

$$\therefore P(A \cup B \cup C \cup D) = 4 \frac{{}^{13}C_2}{{}^{52}C_2} = \frac{4}{17}.$$

Problem 3:

When A and B are mutually exclusive events such that $P(A) = 1/2$ and $P(B) = 1/3$, find $P(A \cup B)$ and $P(A \cap B)$.

Solution :

$$P(A \cup B) = P(A) + P(B) = 5/6 ; P(A \cap B) = 0.$$

Problem 4:

If $P(A) = 0.29$, $P(B) = 0.43$, find $P(A \cap \bar{B})$, if A and B are mutually exclusive.

Solution :

We know $A \cap \bar{B} = A$

$$P(A \cap \bar{B}) = P(A) = 0.29$$

Problem 5:

A card is drawn from a well-shuffled pack of playing cards. What is the probability that it is either a spade or an ace?

Solution :

Let A be the event of drawing a spade

B be the event of drawing an ace

$$\begin{aligned}P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= \frac{13}{52} + \frac{4}{52} - \frac{1}{52} = \frac{4}{13}.\end{aligned}$$

Problem 6:

If $P(A) = 0.4$, $P(B) = 0.7$ and $P(A \cap B) = 0.3$, find $P(\bar{A} \cap \bar{B})$.

Solution :

$$\begin{aligned} P(\bar{A} \cap \bar{B}) &= 1 - P(A \cup B) \\ &= 1 - [P(A) + P(B) - P(A \cap B)] \\ &= 0.2 \end{aligned}$$

Problem 7:

If $P(A) = 0.35$, $P(B) = 0.75$ and $P(A \cup B) = 0.95$, find $P(\bar{A} \cup \bar{B})$.

Solution :

$$P(\bar{A} \cup \bar{B}) = 1 - P(A \cap B) = 1 - [P(A) + P(B) - P(A \cup B)] = 0.85$$

Problem 8:

A lot consists of 10 good articles, 4 with minor defects and 2 with major defects. Two articles are chosen from the lot at random (with out replacement). Find the probability that (i) both are good, (ii) both have major defects, (iii) at least 1 is good, (iv) at most 1 is good, (v) exactly 1 is good, (vi) neither has major defects and (vii) neither is good.

Solution :

$$\begin{aligned} \text{(i)} \quad P(\text{both are good}) &= \frac{{}^{10}C_2}{{}^{16}C_2} = \frac{3}{8} \\ \text{(ii)} \quad P(\text{both have major defects}) &= \frac{{}^2C_2}{{}^{16}C_2} = \frac{1}{120} \\ \text{(iii)} \quad P(\text{at least 1 is good}) &= \frac{{}^{10}C_1 {}^{6}C_1 + {}^{10}C_2}{{}^{16}C_2} = \frac{7}{8} \\ \text{(iv)} \quad P(\text{at most 1 is good}) &= \frac{{}^{10}C_0 {}^6C_2 + {}^{10}C_1 {}^6C_1}{{}^{16}C_2} = \frac{5}{8} \\ \text{(v)} \quad P(\text{exactly 1 is good}) &= \frac{{}^{10}C_1 {}^6C_1}{{}^{16}C_2} = \frac{1}{2} \\ \text{(vi)} \quad P(\text{neither has major defects}) &= \frac{{}^{14}C_2}{{}^{16}C_2} = \frac{91}{120} \\ \text{(vii)} \quad P(\text{neither is good}) &= \frac{{}^6C_2}{{}^{16}C_2} = \frac{1}{8}. \end{aligned}$$

Problem 9:

If A, B and C are any 3 events such that $P(A) = P(B) = P(C) = 1/4$, $P(A \cap B) = P(B \cap C) = 0$; $P(C \cap A) = 1/8$. Find the probability that at least 1 of the events A, B and C occurs.

Solution :

$$\begin{aligned} \text{Since } P(A \cap B) &= P(B \cap C) = 0; P(A \cap B \cap C) = 0 \\ P(A \cup B \cup C) &= P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C) \\ &= \frac{3}{4} - 0 - 0 - \frac{1}{8} = \frac{5}{8}. \end{aligned}$$

Problem 10:

A box contains 4 bad and 6 good tubes. Two are drawn out from the box at a time. One of them is tested and found to be good. What is the probability that the other one is also good?

Solution :

Let A be a good tube drawn and B be an other good tube drawn.

$$P(\text{both tubes drawn are good}) = P(A \cap B) = \frac{{}^6C_2}{{}^{10}C_2} = \frac{1}{3}$$

$$P(B/A) = \frac{P(A \cap B)}{P(A)} = \frac{1/3}{6/10} = \frac{5}{9} \quad (\text{By } \text{conditional probability})$$

Problem 11:

In shooting test, the probability of hitting the target is $1/2$, for A, $2/3$ for B and $3/4$ for C. If all of them fire at the target, find the probability that (i) none of them hits the target and (ii) at least one of them hits the target.

Solution :

Let A, B and C be the event of hitting the target .

$$P(A) = 1/2, P(B) = 2/3, P(C) = 3/4$$

$$P(\bar{A}) = 1/2, P(\bar{B}) = 1/3, P(\bar{C}) = 1/4$$

$$P(\text{none of them hits}) = P(\bar{A} \cap \bar{B} \cap \bar{C}) = P(\bar{A}) \times P(\bar{B}) \times P(\bar{C}) = 1/24$$

$$P(\text{at least one hits}) = 1 - P(\text{none of them hits})$$

$$= 1 - (1/24) = 23/24.$$

Problem 12:

A and B alternatively throw a pair of dice. A wins if he throws 6 before B throws 7 and B wins if he throws 7 before A throws 6. If A begins, show that his chance of winning is $30/61$.

Solution :

Let A be the event of throwing 6

B be the event of throwing 7.

$$P(\text{throwing 6 with 2 dice}) = 5/36$$

$$P(\text{throwing 7 with 2 dice}) = 1/6$$

$$P(\text{not throwing 6}) = 31/36$$

$$P(\text{not throwing 7}) = 5/6$$

A plays in I, III, V,.....trials.

A wins if he throws 6 before B throws 7.

$$P(\text{A wins}) = P(A \cup \bar{A} \bar{B} A \cup \bar{A} \bar{B} \bar{A} \bar{B} A \cup \dots) \\ = P(A) + P(\bar{A} \bar{B} A) + P(\bar{A} \bar{B} \bar{A} \bar{B} A) + \dots$$

$$= \frac{5}{36} + \left(\frac{31}{36} \times \frac{5}{6}\right) \frac{5}{36} + \left(\frac{31}{36} \times \frac{5}{6}\right)^2 \frac{5}{36} + \dots$$

$$= \frac{30}{61}$$

Problem 13:

A and B toss a fair coin alternatively with the understanding that the first who obtain the head wins. If A starts, what is his chance of winning?

Solution :

$$P(\text{getting head}) = 1/2, \quad P(\text{not getting head}) = 1/2$$

A plays in I, III, V,.....trials.

A wins if he gets head before B.

$$\begin{aligned} P(\text{A wins}) &= P(A \cup \bar{A} B A \cup \bar{A} B \bar{A} B A \cup \dots) \\ &= P(A) + P(\bar{A} B A) + P(\bar{A} B \bar{A} B A) + \dots \\ &= \frac{1}{2} + \left(\frac{1}{2} \times \frac{1}{2}\right) \frac{1}{2} + \left(\frac{1}{2} \times \frac{1}{2}\right)^2 \frac{1}{2} + \dots \\ &= \frac{2}{3} \end{aligned}$$

Problem 14:

A problem is given to 3 students whose chances of solving it are $1/2$, $1/3$ and $1/4$. What is the probability that (i) only one of them solves the problem and (ii) the problem is solved.

Solution :

$$P(\text{A solves}) = 1/2 \quad P(B) = 1/3 \quad P(C) = 1/4$$

$$P(\bar{A}) = 1/2, \quad P(\bar{B}) = 2/3, \quad P(\bar{C}) = 3/4$$

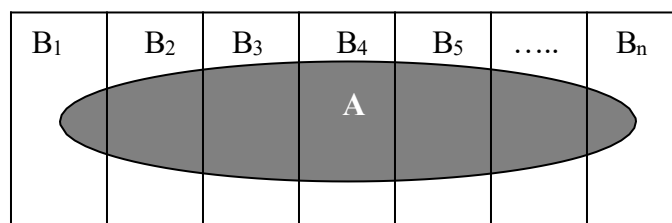
$$P(\text{none of them solves}) = P(\bar{A} \cap \bar{B} \cap \bar{C}) = P(\bar{A}) \times P(\bar{B}) \times P(\bar{C}) = 1/4$$

$$\begin{aligned} P(\text{at least one solves}) &= 1 - P(\text{none of them solves}) \\ &= 1 - (1/4) = 3/4. \end{aligned}$$

Baye's Theorem

Statement: If $B_1, B_2, B_3, \dots, B_n$ be a set of exhaustive and mutually exclusive events associated with a random experiment and A is another event associated with B_i , then

$$P(B_i / A) = \frac{P(B_i) \times P(A / B_i)}{\sum_{i=1}^n P(B_i) \times P(A / B_i)}$$

Proof :

The shaded region represents the event A , A can occur along with $B_1, B_2, B_3, \dots, B_n$ that are mutually exclusive.

$\therefore AB_1, AB_2, AB_3, \dots, AB_n$ are also **mutually exclusive**.

$$\text{Also } A = AB_1 \cup AB_2 \cup AB_3 \cup \dots \cup AB_n$$

$$P(A) = P(AB_1) + P(AB_2) + P(AB_3) + \dots + P(AB_n)$$

$$= \sum_{i=1}^n P(AB_i)$$

$$= \sum_{i=1}^n P(B_i) \times P(A / B_i) \quad (\text{By } \text{conditional probability})$$

$$P(B_i/A) = \frac{P(B_i) \times P(A/B_i)}{P(A)} = \frac{P(B_i) \times P(A/B_i)}{\sum_{i=1}^n P(B_i) \times P(A/B_i)} .$$

Problem 15:

In a bolt factory machines A, B, C manufacture respectively 25%, 35% and 40% of the total. Of their output 5%, 4% and 2% are defective bolts. A bolt is drawn at random from the produce and is found to be defective. What are the probabilities that it was manufactured by machines A, B and C.

Solution :

Let B_1 be bolt produced by machine A

B_2 be bolt produced by machine B

B_3 be bolt produced by machine C

Let A/B_1 be the defective bolts drawn from machine A

A/B_2 be the defective bolts drawn from machine B

A/B_3 be the defective bolts drawn from machine C.

$$P(B_1) = 0.25 \quad P(A/B_1) = 0.05$$

$$P(B_2) = 0.35 \quad P(A/B_2) = 0.04$$

$$P(B_3) = 0.40 \quad P(A/B_3) = 0.02$$

Let B_1/A be defective bolts manufactured by machine A

B_2/A be defective bolts manufactured by machine B

B_3/A be defective bolts manufactured by machine C

$$P(A) = \sum_{i=1}^3 P(B_i) \times P(A/B_i) = (0.25) \times (0.05) + (0.35) \times (0.04) + (0.4) \times (0.02) = 0.0345$$

$$P(B_1/A) = \frac{P(B_1) \times P(A/B_1)}{P(A)} = 0.3623$$

$$P(B_2/A) = \frac{P(B_2) \times P(A/B_2)}{P(A)} = 0.405$$

$$P(B_3/A) = \frac{P(B_3) \times P(A/B_3)}{P(A)} = 0.231 .$$

Problem 16 :

The first bag contains 3 white balls, 2 red balls and 4 black balls. Second bag contains 2 white, 3 red and 5 black balls and third bag contains 3 white, 4 red and 2 black balls. One bag is chosen at random and from it 3 balls are drawn. Out of three balls two balls are white and one is red. What are the probabilities that they were taken from first bag, second bag and third bag.

Solution :

Let $P(\text{selecting the bag}) = P(A_i) = 1/3, i = 1, 2, 3.$

$$P(A/B_1) = \frac{{}^3C_2 {}^2C_1}{{}^9C_3} = \frac{6}{84} \quad P(A) = \sum_{i=1}^3 P(B_i) \times P(A/B_i) = 0.0746$$

$$P(A/B_2) = \frac{{}^2C_2 {}^3C_1}{{}^{10}C_3} = \frac{3}{120}$$

$$P(A/B_3) = \frac{3C_2 4C_1}{9C_3} = \frac{12}{84}$$

$$P(B_1/A) = \frac{P(B_1) \times P(A/B_1)}{P(A)} = 0.319$$

$$P(B_2/A) = \frac{P(B_2) \times P(A/B_2)}{P(A)} = 0.4285$$

$$P(B_3/A) = \frac{P(B_3) \times P(A/B_3)}{P(A)} = 0.638$$

Random Variable

Random Variable:

A random variable is a real valued function whose domain is the sample space of a random experiment taking values on the real line $\mathbb{R}$.

Discrete Random Variable:

A discrete random variable is one which can take only finite or countable number of values with definite probabilities associated with each one of them.

Probability mass function:

Let X be discrete random variable which assuming values $x_1, x_2, \dots, x_n$ with each of the values, we associate a number called the probability $P(X = x_i) = p(x_i), (i = 1, 2, \dots, n)$ this is called the probability of x_i satisfying the following conditions

- i. $p_i \geq 0 \forall i$ i.e., p_i 's are all non-negative
- ii. $\sum_{i=1}^n p_i = p_1 + p_2 + \dots + p_n = 1$ i.e., the total probability is one.

Continuous random variable:

A continuous random variable is one which can assume every value between two specified values with a definite probability associated with each.

Probability Density Function:

A function f is said to be the probability density function of a continuous random variable X if it satisfies the following properties.

- i. $f(x) \geq 0; -\infty < x < \infty$
- ii. $\int_{-\infty}^{\infty} f(x) dx = 1.$

Distribution Function or Cumulative Distribution Function

i. Discrete Variable:

A distribution function of a discrete random variable X is defined as $P(X \leq x) = \sum_{x_i \leq x} P(x_i).$

ii. Continuous Variable:

A distribution function of a continuous random variable X is defined as $F(x) = P(X \leq x) = \int_{-\infty}^x f(x) dx.$

Mathematical Expectation

The expected value of the random variable X is defined as

- i. If X is discrete random variable $E(X) = \sum_{i=1}^{\infty} x_i p(x_i)$ where $p(x)$ is the probability function of x .

- ii. If X is continuous random variable $E(X) = \int_{-\infty}^{\infty} xf(x)dx$ where $f(x)$ is the probability density function of x .

Properties of Expectation:

1. If C is constant then $E(C) = C$

Proof:

Let X be a discrete random variable then $E(x) = \sum xp(x)$

Now $E(C) = \sum Cp(x)$

$$= C \sum p(x) \quad \text{since } \sum_{i=1}^n p_i = p_1 + p_2 + \dots + p_n = 1$$

$$= C$$

2. If a, b are constants then $E(ax + b) = aE(x) + b$

Proof:

Let X be a discrete random variable then $E(x) = \sum xp(x)$

Now $E(ax + b) = \sum (ax + b)p(x)$

$$= \sum axp(x) + \sum bp(x)$$

$$= a \sum xp(x) + b \sum p(x) \quad \text{since } \sum_{i=1}^n p_i = p_1 + p_2 + \dots + p_n = 1$$

$$= aE(x) + b$$

3. If a and b are constants then $Var(ax + b) = a^2Var(x)$

Proof:

$$Var(ax + b) = E[(ax + b - E(ax + b))^2]$$

$$= E[(ax + b - aE(x) - b)^2]$$

$$= E[a^2(x - E(x))^2]$$

$$= a^2E[(x - E(x))^2]$$

$$= a^2Var(x).$$

4. If a is constant then $Var(ax) = a^2Var(x)$

Proof:

$$Var(ax) = E[(ax - E(ax))^2]$$

$$= E[(ax - aE(x))^2]$$

$$= E[a^2(x - E(x))^2]$$

$$= a^2E[(x - E(x))^2]$$

$$= a^2Var(x).$$

5. Prove that $Var(x) = E(x^2) - [E(x)]^2$

Proof:

$$\begin{aligned} Var(x) &= E[(x - E(x))^2] \\ &= E[x^2 + (E(x))^2 - 2xE(x)] \\ &= E[x^2 + \mu^2 - 2x\mu] \\ &= E(x^2) + E(\mu^2) - E(2x\mu) \\ &= E(x^2) + \mu^2 - 2\mu E(x) \\ &= E(x^2) + \mu^2 - 2\mu^2 \\ &= E(x^2) - \mu^2 \end{aligned}$$

$$Var(x) = E(x^2) - [E(x)]^2$$

Problem.1

If the probability distribution of X is given as

$$\begin{array}{l} X : 1 \quad 2 \quad 3 \quad 4 \\ P(X) : 0.4 \quad 0.3 \quad 0.2 \quad 0.1 \end{array}$$

Find $P(1/2 < X < 7/2 / X > 1)$

Solution:

$$\begin{aligned} P\{1/2 < X < 7/2 / X > 1\} &= \frac{P\{(1/2 < X < 7/2) \cap X > 1\}}{P(X > 1)} \\ &= \frac{P(X = 2 \text{ or } 3)}{P(X = 2, 3 \text{ or } 4)} \\ &= \frac{P(X = 2) + P(X = 3)}{P(X = 2) + P(X = 3) + P(X = 4)} \\ &= \frac{0.3 + 0.2}{0.3 + 0.2 + 0.1} = \frac{0.5}{0.6} = \frac{5}{6} \end{aligned}$$

Problem.2

A random variable X has the following probability distribution

$$\begin{array}{l} X : -2 \quad -1 \quad 0 \quad 1 \quad 2 \quad 3 \\ P(X) : 0.1 \quad K \quad 0.2 \quad 2K \quad 0.3 \quad 3K \end{array}$$

a) Find K , b) Evaluate $P(X < 2)$ and $P(-2 < X < 2)$

b) Find the cdf of X and d) Evaluate the mean of X .

Solution:

a) Since $\sum P(X) = 1$

$$\begin{aligned} 0.1 + K + 0.2 + 2K + 0.3 + 3K &= 1 \\ 6K + 0.6 &= 1 \\ 6K &= 0.4 \\ K &= \frac{0.4}{6} = \frac{1}{15} \end{aligned}$$

b) $P(X < 2) = P(X = -2, -1, 0 \text{ or } 1)$

$$\begin{aligned} &= P(X = -2) + P(X = -1) + P(X = 0) + P(X = 1) \\ &= \frac{1}{10} + \frac{1}{15} + \frac{1}{5} + \frac{2}{15} \\ &= \frac{3+2+6+4}{30} = \frac{15}{30} = \frac{1}{2} \end{aligned}$$

$P(-2 < X < 2) = P(X = -1, 0 \text{ or } 1)$

$$= P(X = -1) + P(X = 0) + P(X = 1)$$

$$= \frac{1}{15} + \frac{1}{5} + \frac{2}{15}$$

$$= \frac{1+3+2}{15} = \frac{6}{15} = \frac{2}{5}$$

c) The distribution function of X is given by $F(x)$ defined by

$X = x$	$P(X = x)$	$F(x) = P(X \leq x)$
-2	$\frac{1}{10}$	$F(x) = P(X \leq -2) = \frac{1}{10}$
-1	$\frac{1}{15}$	$F(x) = P(X \leq -1) = \frac{1}{6}$
0	$\frac{2}{10}$	$F(x) = P(X \leq 0) = \frac{11}{30}$
1	$\frac{2}{15}$	$F(x) = P(X \leq 1) = \frac{1}{2}$
2	$\frac{3}{10}$	$F(x) = P(X \leq 2) = \frac{4}{5}$
3	$\frac{3}{15}$	$F(x) = P(X \leq 3) = 1$

d) Mean of X is defined by $E(X)=\sum xP(x)$

$$E(X)=\left(-2\times\frac{1}{10}\right)+\left(-1\times\frac{1}{15}\right)+\left(0\times\frac{1}{5}\right)+\left(1\times\frac{2}{15}\right)+\left(2\times\frac{3}{10}\right)+\left(3\times\frac{1}{5}\right)$$

$$=-\frac{1}{5}-\frac{1}{15}+\frac{2}{15}+\frac{3}{5}+\frac{3}{5}=\frac{16}{15}.$$

Problem.3

A random variable X has the following probability function:

$$\begin{array}{lcl} X & : & 0 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6 \quad 7 \\ P\ X & : & 0 \quad K \quad 2K \quad 2K \quad 3K \quad K^2 \quad 2K^2 \quad 7K^2 + K \end{array}$$

- Find (i) K , (ii) Evaluate $P(X < 6), P(X \geq 6)$ and $P(0 < X < 5)$
- (iii) Determine the distribution function of X .
- (iv) $P(1.5 < X < 4.5/X > 2)$
- (v) $E(3x - 4), Var(3x - 4)$
- (vi). The smallest value of n for which $P(X \leq n) > \frac{1}{2}$.

Solution:

(i) Since $\sum_{x=0}^7 P(X)=1,$

$$K + 2K + 2K + 3K + K^2 + 2K^2 + 7K^2 + K = 1$$

$$10K^2 + 9K - 1 = 0$$

$$K = \frac{1}{10} \text{ or } K = -1$$

As $P(X)$ cannot be negative $K = \frac{1}{10}$

$$\text{(ii)} \quad P(X < 6) = P(X = 0) + P(X = 1) + \dots + P(X = 5) \\ = \frac{1}{10} + \frac{2}{10} + \frac{2}{10} + \frac{3}{10} + \frac{1}{100} + \dots = \frac{81}{100}$$

$$\text{Now } P(X \geq 6) = 1 - P(X < 6) \\ = 1 - \frac{81}{100} = \frac{19}{100}$$

$$\text{Now } P(0 < X < 5) = P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) \\ = K + 2K + 2K + 3K \\ = 8K = \frac{8}{10} = \frac{4}{5}$$

(iii) The distribution of X is given by $F(x) = P(X \leq x)$

$X = x$	$P(X = x)$	$F(x) = P(X \leq x)$
0	0	$F(x) = P(X \leq 0) = 0$
1	$\frac{1}{10}$	$F(x) = P(X \leq 1) = \frac{1}{10}$
2	$\frac{2}{10}$	$F(x) = P(X \leq 2) = \frac{3}{10}$
3	$\frac{2}{10}$	$F(x) = P(X \leq 3) = \frac{5}{10}$
4	$\frac{3}{10}$	$F(x) = P(X \leq 4) = \frac{8}{10}$
5	$\frac{1}{100}$	$F(x) = P(X \leq 5) = \frac{81}{100}$
6	$\frac{2}{100}$	$F(x) = P(X \leq 6) = \frac{83}{100}$
7	$\frac{17}{100}$	$F(x) = P(X \leq 7) = 1$

$$\text{(iv)} \quad P(1.5 < X < 4.5 / X > 2) = \frac{P(x=3) + P(x=4)}{1 - [P(x=0) + P(x=1) + P(x=2)]} \\ = \frac{\frac{5}{10}}{1 - [\frac{3}{10}]} = \frac{5}{7}$$

$$\text{(v)} \quad E(x) = \sum xp(x) \\ = 1 \times \frac{1}{10} + 2 \times \frac{2}{10} + 3 \times \frac{2}{10} + 4 \times \frac{3}{10} + 5 \times \frac{1}{100} + 6 \times \frac{2}{100} + 7 \times \frac{17}{100} \\ E(x) = 3.66$$

$$E(x^2) = \sum x^2 p(x)$$

$$= 1^2 \times \frac{1}{10} + 2^2 \times \frac{2}{10} + 3^2 \times \frac{2}{10} + 4^2 \times \frac{3}{10} + 5^2 \times \frac{1}{100} + 6^2 \times \frac{2}{100} + 7^2 \times \frac{17}{100}$$

$$E(x^2) = 16.8$$

$$\text{Mean} = E(x) = 3.66$$

$$\text{Variance} = E(x^2) - [E(x)]^2$$

$$= 16.8 - (3.66)^2$$

$$= 3.404$$

(vi) The smallest value of n for which $P(X \leq n) > \frac{1}{2}$ is 4

Problem.4

The probability mass function of random variable X is defined as $P(X=0) = 3C^2$, $P(X=1) = 4C - 10C^2$, $P(X=2) = 5C - 1$, where $C > 0$, and $P(X=r) = 0$ if $r \neq 0, 1, 2$.

Find (i). The value of C .

(ii) $P(0 < X < 2/x > 0)$.

(iii) The distribution function of X .

(iv). The largest value of x for which $F(x) < \frac{1}{2}$.

Solution:

(i) Since $\sum_{x=0}^{\infty} p(x) = 1$

$$p(0) + p(1) + p(2) = 1$$

$$3C^2 + 4C - 10C^2 + 5C - 1 = 1$$

$$7C^2 - 9C + 2 = 0$$

$$C = 1, \frac{2}{7}$$

$C = 1$ is not applicable

$$\therefore C = \frac{2}{7}$$

The Probability distribution is

$$\begin{array}{lcl} X & : & 0 \quad 1 \quad 2 \\ P(X) & : & \frac{12}{49} \quad \frac{16}{49} \quad \frac{21}{49} \end{array}$$

$$(ii) P\left[0 < x < 2/x > 0\right] = \frac{P[(0 < x < 2) \cap x > 0]}{P[x > 0]}$$

$$= \frac{P[0 < x < 2]}{P[x > 0]} = \frac{P[x=1]}{P[x=1] + P[X=2]}$$

$$P\left[0 < x < 2/x > 0\right] = \frac{\frac{16}{49}}{\frac{16}{49} + \frac{21}{49}} = \frac{16}{37}$$

(iii). The distribution function of X is

X	$F(X=x)=P(X\leq x)$
0	$F(0)=P(X\leq 0)=\frac{12}{49}=0.24$
1	$F(1)=P(X\leq 1)=P(X=0)+P(X=1)=\frac{12}{49}+\frac{16}{49}=0.57$
2	$F(2)=P(X\leq 2)=P(X=0)+P(X=1)+P(X=2)=\frac{12}{49}+\frac{16}{49}+\frac{21}{49}=1$

(iv) The Largest value of x for which $F(x)=P(X\leq x)<\frac{1}{2}$ is 0.

Problem.5

$$\text{If } P(x) = \begin{cases} \frac{x}{15}; & x=1,2,3,4,5 \\ 0 & ; \text{elsewhere} \end{cases}$$

Find (i) $P\{X=1 \text{ or } 2\}$ and (ii) $P\{1/2 < X < 5/2 / x > 1\}$

Solution:

$$\text{i) } P(X=1 \text{ or } 2) = P(X=1) + P(X=2)$$

$$= \frac{1}{15} + \frac{2}{15} = \frac{3}{15} = \frac{1}{5}$$

$$\text{ii) } P\left(\frac{1}{2} < X < \frac{5}{2} / x > 1\right) = \frac{P\left\{\left(\frac{1}{2} < X < \frac{5}{2}\right) \cap (X > 1)\right\}}{P(X > 1)}$$

$$= \frac{P\{(X=1 \text{ or } 2) \cap (X > 1)\}}{P(X > 1)}$$

$$= \frac{P(X=2)}{1 - P(X=1)}$$

$$= \frac{2/15}{1 - (1/15)} = \frac{2/15}{14/15} = \frac{2}{14} = \frac{1}{7}.$$

Problem.6

A continuous random variable X has a probability density function $f(x) = 3x^2$, $0 \leq x \leq 1$. Find 'a' such that $P(X \leq a) = P(X > a)$.

Solution:

Since $P(X \leq a) = P(X > a)$, each must be equal to $\frac{1}{2}$ because the probability is always 1.

$$\therefore P(X \leq a) = \frac{1}{2}$$

$$\Rightarrow \int_0^a f(x) dx = \frac{1}{2}$$

$$\int_0^a 3x^2 dx = \frac{1}{2} \Rightarrow 3 \left[\frac{x^3}{3} \right]_0^a = \frac{1}{2}.$$

$$\therefore a = \left(\frac{1}{2}\right)^{\frac{1}{3}}$$

Problem.7

A random variable X has the p.d.f $f(x)$ given by $f(x) = \begin{cases} Cxe^{-x}; & \text{if } x > 0 \\ 0 & ; \text{if } x \leq 0 \end{cases}$ Find the value of C and cumulative density function of X .

Solution:

Since $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\int_0^{\infty} Cxe^{-x} dx = 1$$

$$C \left[x(-e^{-x}) - (e^{-x}) \right]_0^{\infty} = 1$$

$$C = 1$$

$$\therefore f(x) = \begin{cases} xe^{-x}; & x > 0 \\ 0 & ; x \leq 0 \end{cases}$$

Cumulative Distribution of x is

$$\begin{aligned} F(x) &= \int_0^x f(x) dt = \int_0^x xe^{-x} dx = \left[-xe^{-x} - e^{-x} \right]_0^x = -xe^{-x} - e^{-x} + 1 \\ &= 1 - (1+x)e^{-x}, \quad x > 0. \end{aligned}$$

Problem.8

If a random variable X has the p.d.f $f(x) = \begin{cases} \frac{1}{2}(x+1); & -1 < x < 1 \\ 0 & ; \text{otherwise} \end{cases}$. Find the mean and

variance of X .

Solution:

$$\text{Mean} = \mu_1' = \int_{-1}^1 xf(x) dx = \frac{1}{2} \int_{-1}^1 x(x+1) dx = \frac{1}{2} \int_{-1}^1 (x^2 + x) dx$$

$$= \frac{1}{2} \left(\frac{x^3}{3} + \frac{x^2}{2} \right)_{-1}^1 = \frac{1}{3}$$

$$\mu_2' = \int_{-1}^1 x^2 f(x) dx = \frac{1}{2} \int_{-1}^1 (x^3 + x^2) dx = \frac{1}{2} \left[\frac{x^4}{4} + \frac{x^3}{3} \right]_{-1}^1$$

$$= \frac{1}{2} \left[\frac{1}{4} + \frac{1}{3} - \frac{1}{4} + \frac{1}{3} \right]$$

$$= \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}$$

$$\text{Variance} = \mu_2' - (\mu_1')^2$$

$$= \frac{1}{3} - \frac{1}{9} = \frac{3-1}{9} = \frac{2}{9}.$$

Problem.9

A continuous random variable X that can assume any value between $X = 2$ and $X = 5$ has a probability density function given by $f(x) = k(1+x)$. Find $P(X < 4)$.

Solution:

Given X is a continuous random variable whose pdf is $f(x) = \begin{cases} k(1+x), & 2 < x < 5 \\ 0, & \text{Otherwise} \end{cases}$.

$$\text{Since } \int_{-\infty}^{\infty} f(x) dx = 1 \Rightarrow \int_2^5 k(1+x) dx = 1$$

$$k \left[\frac{(1+x)^2}{2} \right]_2^5 = 1$$

$$k \left[\frac{(1+5)^2}{2} - \frac{(1+2)^2}{2} \right] = 1$$

$$k \left[18 - \frac{9}{2} \right] = 1$$

$$k \left[\frac{27}{2} \right] = 1 \Rightarrow k = \frac{2}{27}$$

$$\therefore f(x) = \begin{cases} \frac{2(1+x)}{27}, & 2 < x < 5 \\ 0, & \text{Otherwise} \end{cases}$$

$$\begin{aligned} P(X < 4) &= \frac{2}{27} \int_2^4 (1+x) dx \\ &= \frac{2}{27} \left[\frac{(1+x)^2}{2} \right]_2^4 = \frac{2}{27} \left[\frac{(1+4)^2}{2} - \frac{(1+2)^2}{2} \right] = \frac{2}{27} \left[\frac{25}{2} - \frac{9}{2} \right] = \frac{2}{27} \frac{16}{2} = \frac{16}{27}. \end{aligned}$$

Problem.10

A random variable X has density function given by $f(x) = \begin{cases} 2e^{-2x}; & x \geq 0 \\ 0 & ; x < 0 \end{cases}$. Find m.g.f

Solution:

$$\begin{aligned} M_X(t) &= E(e^{tx}) = \int_0^{\infty} e^{tx} f(x) dx = \int_0^{\infty} e^{tx} 2e^{-2x} dx \\ &= 2 \int_0^{\infty} e^{(t-2)x} dx \\ &= 2 \left[\frac{e^{(t-2)x}}{t-2} \right]_0^{\infty} = \frac{2}{2-t}, t < 2. \end{aligned}$$

Problem.11

The pdf of a random variable X is given by $f(x) = \begin{cases} 2x, & 0 \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$. For what value of b is

$f(x)$ a valid pdf? Also find the cdf of the random variable X with the above pdf.

Solution:

$$\text{Given } f(x) = \begin{cases} 2x, & 0 \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$$

$$\text{Since } \int_{-\infty}^{\infty} f(x) dx = 1 \Rightarrow \int_0^b 2x dx = 1$$

$$\left[2 \frac{x^2}{2} \right]_0^b = 1$$

$$\left[b^2 - 0 \right] = 1 \Rightarrow b = 1$$

$$\therefore f(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

$$F(x) = P(X \leq x) = \int_0^x f(x) dx = \int_0^x 2x dx = \left[2 \frac{x^2}{2} \right]_0^x = x^2, \quad 0 \leq x \leq 1$$

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(x) dx = \int_{-\infty}^x 0 dx = 0, \quad x < 0$$

$$\begin{aligned} F(x) = P(X \leq x) &= \int_{-\infty}^0 f(x) dx + \int_0^1 f(x) dx + \int_1^x f(x) dx \\ &= \int_{-\infty}^0 0 dx + \int_0^1 2x dx + \int_1^x 0 dx = \left[2 \frac{x^2}{2} \right]_0^1 = 1, \quad x > 1 \end{aligned}$$

$$F(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 \leq x \leq 1 \\ 1, & x > 1 \end{cases}$$

Problem.12

A random variable X has density function $f(x) = \begin{cases} \frac{K}{1+x^2}, & -\infty < x < \infty \\ 0, & \text{Otherwise} \end{cases}$. Determine K

and the distribution functions. Evaluate the probability $P(x \geq 0)$.

Solution:

$$\text{Since } \int_{-\infty}^{\infty} f(x) dx = 1$$

$$\int_{-\infty}^{\infty} \frac{K}{1+x^2} dx = 1$$

$$K \int_{-\infty}^{\infty} \frac{dx}{1+x^2} = 1$$

$$K \left(\tan^{-1} x \right)_{-\infty}^{\infty} = 1$$

$$K \left(\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right) = 1$$

$$K\pi = 1$$

$$\begin{aligned}
 K &= \frac{1}{\pi} \\
 F(x) &= \int_{-\infty}^x f(x)dx = \int_{-\infty}^x \frac{K}{1+x^2} dx \\
 &= \frac{1}{\pi} \left[\tan^{-1} x - \left(-\frac{\pi}{2} \right) \right] \\
 F(x) &= \frac{1}{\pi} \left[\frac{\pi}{2} + \tan^{-1} x \right]_{-\infty < x < \infty} \\
 P(X \geq 0) &= \frac{1}{\pi} \int_0^{\infty} \frac{dx}{1+x^2} = \frac{1}{\pi} \left(\tan^{-1} x \right)_0^{\infty} \\
 &= \frac{1}{\pi} \left(\frac{\pi}{2} - \tan^{-1} 0 \right) = \frac{1}{2}.
 \end{aligned}$$

Problem.13

If X has the probability density function $f(x) = \begin{cases} Ke^{-3x}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$ find K ,

$P[0.5 \leq X \leq 1]$ and the mean of X .

Solution:

$$\begin{aligned}
 \text{Since } \int_{-\infty}^{\infty} f(x) dx &= 1 \\
 \int_0^{\infty} Ke^{-3x} dx &= 1 \\
 K \left[\frac{e^{-3x}}{-3} \right]_0^{\infty} &= 1 \\
 \frac{K}{3} &= 1 \\
 K &= 3 \\
 P(0.5 \leq X \leq 1) &= \int_{0.5}^1 f(x) dx = 3 \int_{0.5}^1 e^{-3x} dx = 3 \left(\frac{e^{-3} - e^{-1.5}}{-3} \right) = [e^{-1.5} - e^{-3}] \\
 \text{Mean of } X = E(x) &= \int_0^{\infty} xf(x) dx = 3 \int_0^{\infty} xe^{-3x} dx \\
 &= 3 \left[x \left(\frac{-e^{-3x}}{3} \right) - 1 \left(\frac{e^{-3x}}{9} \right) \right]_0^{\infty} = \frac{3 \times 1}{9} = \frac{1}{3} \\
 \text{Hence the mean of } X = E(X) &= \frac{1}{3}.
 \end{aligned}$$

Problem.14

If X is a continuous random variable with pdf given by

$$f(x) = \begin{cases} Kx & \text{in } 0 \leq x \leq 2 \\ 2K & \text{in } 2 \leq x \leq 4 \\ 6K - Kx & \text{in } 4 \leq x \leq 6 \\ 0 & \text{elsewhere} \end{cases}. \text{ Find the value of } K \text{ and also the cdf } F(x).$$

Solution:

$$\text{Since } \int_{-\infty}^{\infty} f(x) dx = 1$$

$$\int_0^2 Kx dx + \int_2^4 2K dx + \int_4^6 (6K - Kx) dx = 1$$

$$K \left[\left(\frac{x^2}{2} \right)_0^2 + (2x)_2^4 + \left(6x - \frac{x^2}{2} \right)_4^6 \right] = 1$$

$$K [2 + 8 - 4 + 36 - 18 - 24 + 8] = 1$$

$$8K = 1$$

$$K = \frac{1}{8}$$

$$\text{We know that } F(x) = \int_{-\infty}^x f(x) dx$$

$$\text{If } x < 0, \text{ then } F(x) = \int_{-\infty}^x f(x) dx = 0$$

$$\text{If } x \in (0, 2), \text{ then } F(x) = \int_{-\infty}^x f(x) dx$$

$$F(x) = \int_{-\infty}^0 f(x) dx + \int_0^x f(x) dx$$

$$= \int_{-\infty}^0 0 dx + \int_0^x Kx dx = \int_{-\infty}^0 0 dx + \frac{1}{8} \int_0^x x dx$$

$$F(x) = \left(\frac{x^2}{16} \right)_0^x = \frac{x^2}{16}, 0 \leq x \leq 2$$

$$\text{If } x \in (2, 4), \text{ then } F(x) = \int_{-\infty}^0 f(x) dx + \int_0^2 f(x) dx + \int_2^x f(x) dx$$

$$= \int_{-\infty}^0 0 dx + \int_0^2 Kx dx + \int_2^x 2K dx$$

$$= \int_0^2 \frac{x}{8} dx + \int_2^x \frac{1}{4} dx = \left(\frac{x^2}{16} \right)_0^2 + \left(\frac{x}{4} \right)_2^x$$

$$= \frac{1}{4} + \frac{x}{4} - \frac{1}{2}$$

$$F(x) = \frac{x}{4} - \frac{4}{16} = \frac{x-1}{4}, 2 \leq x < 4$$

$$\begin{aligned}
\text{If } x \in (4, 6), \text{ then } F(x) &= \int_{-\infty}^0 0 dx + \int_0^2 Kx dx + \int_2^4 2K dx + \int_4^x K(6-x) dx \\
&= \int_0^2 \frac{x}{8} dx + \int_2^4 \frac{1}{4} dx + \int_4^x \frac{1}{8} (6-x) dx \\
&= \left(\frac{x^2}{16} \right)_0^2 + \left(\frac{x}{4} \right)_2^4 + \left(\frac{6x}{8} - \frac{x^2}{16} \right)_4^x \\
&= \frac{1}{4} + 1 - \frac{1}{2} + \frac{6x}{8} - \frac{x^2}{16} - 3 + 1 \\
&= \frac{4 + 16 - 8 + 12x - x^2 - 48 + 16}{16}
\end{aligned}$$

$$F(x) = \frac{-x^2 + 12x - 20}{16}, 4 \leq x \leq 6$$

$$\begin{aligned}
\text{If } x > 6, \text{ then } F(x) &= \int_{-\infty}^0 0 dx + \int_0^2 Kx dx + \int_2^4 2K dx + \int_4^6 K(6-x) dx + \int_6^{\infty} 0 dx \\
F(x) &= 1, x \geq 6
\end{aligned}$$

$$\therefore F(x) = \begin{cases} 0 & ; x \leq 0 \\ \frac{x^2}{16} & ; 0 \leq x \leq 2 \\ \frac{1}{4}(x-1) & ; 2 \leq x \leq 4 \\ \frac{-1}{16}(20-12x+x^2) & ; 4 \leq x \leq 6 \\ 1 & ; x \geq 6 \end{cases}$$

Problem.15

A random variable X has the P.d.f $f(x) = \begin{cases} 2x, 0 < x < 1 \\ 0, \text{ Otherwise} \end{cases}$

$$\text{Find (i) } P\left(X < \frac{1}{2}\right) \text{ (ii) } P\left(\frac{1}{4} < x < \frac{1}{2}\right) \text{ (iii) } P\left(X > \frac{3}{4} / X > \frac{1}{2}\right)$$

Solution:

$$(i) P\left(x < \frac{1}{2}\right) = \int_0^{1/2} f(x) dx = \int_0^{1/2} 2x dx = 2 \left(\frac{x^2}{2} \right)_0^{1/2} = \frac{2 \times 1}{8} = \frac{1}{4}$$

$$\begin{aligned}
(ii) P\left(\frac{1}{4} < x < \frac{1}{2}\right) &= \int_{1/4}^{1/2} f(x) dx = \int_{1/4}^{1/2} 2x dx = 2 \left(\frac{x^2}{2} \right)_{1/4}^{1/2} \\
&= 2 \left(\frac{1}{8} - \frac{1}{32} \right) = \left(\frac{1}{4} - \frac{1}{16} \right) = \frac{3}{16}
\end{aligned}$$

$$(iii) P\left(X > \frac{3}{4} / X > \frac{1}{2}\right) = \frac{P\left(X > \frac{3}{4} \cap X > \frac{1}{2}\right)}{P\left(X > \frac{1}{2}\right)} = \frac{P\left(X > \frac{3}{4}\right)}{P\left(X > \frac{1}{2}\right)}$$

$$P\left(X > \frac{3}{4}\right) = \int_{3/4}^1 f(x) dx = \int_{3/4}^1 2x dx = 2 \left(\frac{x^2}{2} \right)_{3/4}^1 = 1 - \frac{9}{16} = \frac{7}{16}$$

$$P\left(X > \frac{1}{2}\right) = \int_{1/2}^1 f(x) dx = \int_{1/2}^1 2x dx = 2 \left(\frac{x^2}{2} \right)_{1/2}^1 = 1 - \frac{1}{4} = \frac{3}{4}$$

$$P\left(X > \frac{3}{4} / X > \frac{1}{2}\right) = \frac{\frac{7}{16}}{\frac{3}{4}} = \frac{7}{16} \times \frac{4}{3} = \frac{7}{12}.$$

Problem.16

Let the random variable X have the p.d.f $f(x) = \begin{cases} \frac{1}{2} e^{-x/2}, & x > 0 \\ 0, & \text{otherwise.} \end{cases}$ Find the moment

generating function, mean & variance of X .

Solution:

$$M_X(t) = E(e^{tx}) = \int_{-\infty}^{\infty} e^{tx} f(x) dx = \int_0^{\infty} e^{tx} \frac{1}{2} e^{-x/2} dx$$

$$= \frac{1}{2} \int_0^{\infty} e^{-\left(\frac{1}{2}-t\right)x} dx = \frac{1}{2} \left[\frac{e^{-\left(\frac{1}{2}-t\right)x}}{-\left(\frac{1}{2}-t\right)} \right]_0^{\infty} = \frac{1}{1-2t}, \text{ if } t < \frac{1}{2}.$$

$$E(X) = \left[\frac{d}{dt} M_X(t) \right]_{t=0} = \left[\frac{2}{(1-2t)^2} \right]_{t=0} = 2$$

$$E(X^2) = \left[\frac{d^2}{dt^2} M_X(t) \right]_{t=0} = \left[\frac{8}{(1-2t)^3} \right]_{t=0} = 8$$

$$Var(X) = E(X^2) - [E(X)]^2 = 8 - 4 = 4.$$

Probability and Statistics

Probability Distribution

Discrete type

Binomial distribution:

A random variable X is said to follow binomial distribution if it assumes only non negative values and its probability mass function is given by

$$P(X = x) = p(x) = \begin{cases} nC_x p^x q^{n-x}, & x = 0, 1, 2, \dots, n; q = 1 - p \\ 0, & \text{otherwise} \end{cases}$$

Notation: $X \sim B(n, p)$ read as X is following binomial distribution with parameter n and p .

Problem.1

Find m.g.f. of Binomial distribution and find its mean and variance.

Solution:

M.G.F. of Binomial distribution:-

$$\begin{aligned} M_X(t) &= E[e^{tx}] = \sum_{x=0}^n e^{tx} P(X = x) \\ &= \sum_{x=0}^n nC_x p^x q^{n-x} e^{tx} \\ &= \sum_{x=0}^n nC_x (pe^t)^x q^{n-x} \\ M_X(t) &= (q + pe^t)^n \end{aligned}$$

Mean of Binomial distribution

$$\begin{aligned} \text{Mean} &= E(X) = M_X'(0) \\ &= \left[n(q + pe^t)^{n-1} pe^t \right]_{t=0} = np \quad \text{Since } q + p = 1 \\ E(X^2) &= M_X''(0) \\ &= \left[n(n-1)(q + pe^t)^{n-2} (pe^t)^2 + npe^t (q + pe^t)^{n-1} \right]_{t=0} \\ E(X^2) &= n(n-1)p^2 + np \\ &= n^2 p^2 + np(1-p) = n^2 p^2 + npq \end{aligned}$$

$$\text{Variance} = E(X^2) - [E(X)]^2 = npq$$

$$\text{Mean} = np; \text{Variance} = npq$$

Problem.2

Comment the following: "The mean of a binomial distribution is 3 and variance is 4"

Solution:

In binomial distribution, mean > variance but Variance < Mean

Since Variance = 4 & Mean = 3, the given statement is wrong.

Problem.3

If X and Y are independent binomial variates $B\left(5, \frac{1}{2}\right)$ and $B\left(7, \frac{1}{2}\right)$ find $P[X + Y = 3]$

Solution:

$X + Y$ is also a binomial variate with parameters $n_1 + n_2 = 12$ & $p = \frac{1}{2}$

$$\therefore P[X + Y = 3] = {}^{12}C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^9 = \frac{55}{2^{10}}$$

Problem.4

(i). Six dice are thrown 729 times. How many times do you expect atleast 3 dice show 5 or 6 ?

(ii) Six coins are tossed 6400 times. Using the Poisson distribution, what is the approximate probability of getting six heads 10 times?

Solution:

(i). Let X be the number of times the dice shown 5 or 6

$$P[5 \text{ or } 6] = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$$

$$\therefore P = \frac{1}{3} \text{ and } q = \frac{2}{3}$$

Here $n = 6$

By Binomial theorem,

$$P[X = x] = {}^6C_x \left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{6-x} \text{ where } x = 0, 1, 2, \dots, 6.$$

$$\begin{aligned} P[X \geq 3] &= P(3) + P(4) + P(5) + P(6) \\ &= {}^6C_3 \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^3 + {}^6C_4 \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^2 + {}^6C_5 \left(\frac{1}{3}\right)^5 \left(\frac{2}{3}\right) + {}^6C_6 \left(\frac{1}{3}\right)^6 \\ &= 0.3196 \end{aligned}$$

$$\begin{aligned} \therefore \text{Expected number of times atleast 3 dies to show 5 or 6} &= N \times P[X \geq 3] \\ &= 729 \times 0.3196 = 233. \end{aligned}$$

(ii). Probability of getting six heads in one toss of six coins is $p = \left(\frac{1}{2}\right)^6$,

$$\lambda = np = 6400 \times \left(\frac{1}{2}\right)^6 = 100$$

$$\text{Let } X \text{ be the number of times getting 6 heads } P(X=10) = \frac{e^{-100} (100)^{10}}{10!} = 1.025 \times 10^{-30}$$

Poisson distribution:

A random variable X is said to follow Poisson distribution if it assumes only non negative values and its probability mass function is given by

$$P(X = x) = \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!}; x = 0, 1, 2, \dots; \lambda > 0 \\ 0, \text{otherwise} \end{cases}$$

Notation: $X \sim P(\lambda)$ read as X is following Poisson distribution with parameter λ .

Poisson distribution as limiting form of binomial distribution:

Poisson distribution is a limiting case of Binomial distribution under the following conditions:

- (i) n the number of trials is indefinitely large, (i.e.) $n \rightarrow \infty$
- (ii) p the constant probability of success in each trial is very small (i.e.) $p \rightarrow 0$
- (iii) $np = \lambda$ is finite.

Proof:

$$P(X = x) = p(x) = {}^n C_x p^x q^{n-x}$$

Let $np = \lambda$

$$\therefore p = \frac{\lambda}{n}, q = 1 - \frac{\lambda}{n}$$

$$\begin{aligned} \therefore p(x) &= {}^n C_x \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x} \\ &= \frac{n!}{x!(n-x)!} \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x} \\ &= \frac{n(n-1)\cdots(n-(x-1)) \cancel{(n-x)!}}{x! \cancel{(n-x)!}} \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x} \\ &= \frac{1 \cdot \left(1 - \frac{1}{n}\right) \cdots \left(1 - \frac{x-1}{n}\right)}{x!} \cancel{n^x} \frac{\lambda^x}{\cancel{n^x}} \left(1 - \frac{\lambda}{n}\right)^{n-x} \\ p(x) &= 1 \cdot \left(1 - \frac{1}{n}\right) \cdots \left(1 - \frac{x-1}{n}\right) \frac{\lambda^x}{x!} \left(1 - \frac{\lambda}{n}\right)^{n-x} \end{aligned}$$

Taking limit $n \rightarrow \infty$ on both sides

$$\begin{aligned} \lim_{n \rightarrow \infty} p(x) &= \frac{\lambda^x}{x!} \lim_{n \rightarrow \infty} \left[\left(1 - \frac{1}{n}\right) \cdots \left(1 - \frac{x-1}{n}\right) \left(1 - \frac{\lambda}{n}\right)^{n-x} \right] \\ &= \frac{\lambda^x}{x!} \lim_{n \rightarrow \infty} \left[\left(1 - \frac{1}{n}\right) \cdots \left(1 - \frac{x-1}{n}\right) \right] \lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^{-x} \lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^n \end{aligned}$$

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}; x = 0, 1, 2, \dots$$

Problem.1

Criticise the following statement: "The mean of a Poisson distribution is 5 while the standard deviation is 4".

Solution:

For a Poisson distribution mean and variance are same. Hence this statement is not true.

Geometric distribution:

A random variable X is said to have a Geometric distribution if it assumes only non negative values and its probability mass function is given by

$$P(X = x) = \begin{cases} q^{x-1} p; x = 1, 2, \dots; 0 < p \leq 1 \\ 0, \text{otherwise} \end{cases}$$

Problem.1

Find the Moment generating function of geometric distribution and find its Mean and Variance

Solution:

$$\begin{aligned} M_X(t) &= E(e^{tX}) \\ &= \sum_{x=1}^{\infty} e^{tx} q^{x-1} p \\ &= \sum_{x=1}^{\infty} p e^t (q e^t)^{x-1} \\ &= p e^t \left(1 + q e^t + (q e^t)^2 + \dots \right) \\ &= p e^t (1 - q e^t)^{-1} \end{aligned}$$

$$M_X(t) = \frac{p e^t}{1 - q e^t}$$

$$\mu_1' = M_X'(0) = \left[\frac{d}{dt} \left(\frac{p e^t}{(1 - q e^t)} \right) \right]_{t=0} = \left[\left(\frac{p e^t}{(1 - q e^t)^2} \right) \right]_{t=0} = \frac{1}{p}$$

$$\mu_2' = M_X''(0) = \left[\frac{d^2}{dt^2} \left(\frac{p e^t}{(1 - q e^t)} \right) \right]_{t=0} = \left[\frac{d}{dt} \left(\frac{p e^t}{(1 - q e^t)^2} \right) \right]_{t=0} = \frac{1+q}{p^2}$$

$$\text{Mean} = \mu_1' = \frac{1}{p}$$

$$\text{Variance} = \mu_2' - (\mu_1')^2 = \frac{1+q}{p^2} - \left(\frac{1}{p} \right)^2 = \frac{q}{p^2}$$

Problem.2

State and prove Memoryless property of geometric distribution.

Solution:

If X has a geometric distribution, then for any two positive integer 's' and 't'

$$P\left[\frac{X > s+t}{X > s} \right] = P[X > t].$$

The p.m.f of the geometric random variable X is $P(X = x) = q^{x-1} p$, $x = 1, 2, 3, \dots$

$$P\left[\frac{X > s+t}{X > s} \right] = \frac{P[X > s+t \cap X > s]}{P[X > s]} = \frac{P[X > s+t]}{P[X > s]} \text{ -----(1)}$$

$$\begin{aligned}\therefore P[X > t] &= \sum_{x=t+1}^{\infty} q^{x-1} p = q^t p + q^{t+1} p + q^{t+2} p + \dots = q^t p [1 + q + q^2 + q^3 + \dots] \\ &= q^t p (1 - q)^{-1} = q^t p (p)^{-1} = q^t\end{aligned}$$

Hence $P[X > s + t] = q^{s+t}$ and $P[X > s] = q^s$

$$(1) \Rightarrow P\left[X > s + t \middle/ X > s\right] = \frac{P[X > s + t \cap X > s]}{P[X > s]} = \frac{q^{s+t}}{q^s} = q^t = P[X > t]$$

$$\Rightarrow P\left[X > s + t \middle/ X > s\right] = P(X > t)$$

Problem.3

If the probability is $\frac{1}{4}$ that a man will hit a target what is the chance that he will hit the target for the first time in the 7th trial?

Solution:

The required probability is

$$\begin{aligned}P[FFFFFF S] &= P(F)P(F)P(F)P(F)P(F)P(F)P(S) \\ &= q^6 p = \left(\frac{3}{4}\right)^6 \cdot \left(\frac{1}{4}\right) = 0.0445.\end{aligned}$$

Problem.4

A die is cast until 6 appears what is the probability that it must cast more than five times?

Solution:

$$\text{Probability of getting six} = \frac{1}{6}$$

$$\therefore p = \frac{1}{6} \text{ \& } q = 1 - \frac{1}{6}$$

Let x : No of throws for getting the number 6. By geometric distribution $P[X = x] = q^{x-1} p, x = 1, 2, 3, \dots$

Since 6 can be got either in first, second.....throws.

To find $P[X \geq 6] = 1 - P[X < 6]$

$$\begin{aligned}&= 1 - \sum_{x=1}^5 \left(\frac{5}{6}\right)^{x-1} \cdot \frac{1}{6} \\ &= 1 - \left[\left(\frac{1}{6}\right) + \left(\frac{5}{6}\right)\left(\frac{1}{6}\right) + \left(\frac{5}{6}\right)^2 \left(\frac{1}{6}\right) + \left(\frac{5}{6}\right)^3 \left(\frac{1}{6}\right) + \left(\frac{5}{6}\right)^4 \left(\frac{1}{6}\right)\right] \\ &= 1 - \frac{\frac{1}{6} \left[1 - \left(\frac{5}{6}\right)^5\right]}{1 - \frac{5}{6}} = \left(\frac{5}{6}\right)^5 = 0.4019\end{aligned}$$

Problem.5

Suppose that a trainee soldier shoots a target an independent fashion. If the probability that the target is shot on any one shot is 0.8.

(i) What is the probability that the target would be hit on 6th attempt?

(ii) What is the probability that it takes him less than 5 shots?

Solution:

Here $p = 0.8, q = 1 - p = 0.2$

$$P[X = x] = q^{x-1} p, x = 1, 2, \dots$$

(i) The probability that the target would be hit on the 6th attempt = $P[X = 6]$
 $= (0.2)^5 (0.8) = 0.00026$

(ii) The probability that it takes him less than 5 shots = $P[X < 5]$
 $= \sum_{x=1}^4 q^{x-1} p = 0.8 \sum_{x=1}^4 (0.2)^{x-1}$
 $= 0.8 [1 + 0.2 + 0.04 + 0.008] = 0.9984$

Continuous type**Uniform (or) Rectangular distribution:**

A continuous random variable X is said to have a uniform distribution over an interval (a, b) if its probability density function is given by

$$f(x) = \begin{cases} \frac{1}{b-a}, & a < x < b \\ 0, & \text{otherwise} \end{cases}$$

Problem.1

If X is uniformly distributed with Mean 1 and Variance $\frac{4}{3}$, find $P[X > 0]$

Solution:

If X is uniformly distributed over (a, b) , then

$$E(X) = \frac{b+a}{2} \text{ and } V(X) = \frac{(b-a)^2}{12}$$

$$\therefore \frac{b+a}{2} = 1 \Rightarrow a+b = 2$$

$$\Rightarrow \frac{(b-a)^2}{12} = \frac{4}{3} \Rightarrow (b-a)^2 = 16$$

$$\Rightarrow a+b = 2 \text{ \& } b-a = 4 \text{ We get } b = 3, a = -1$$

$\therefore a = -1$ & $b = 3$ and probability density function of x is

$$f(x) = \begin{cases} \frac{1}{4}, & -1 < x < 3 \\ 0, & \text{Otherwise} \end{cases}$$

$$P[x < 0] = \int_{-1}^0 \frac{1}{4} dx = \frac{1}{4} [x]_{-1}^0 = \frac{1}{4}$$

Exponential distribution:

A continuous random variable X assuming non negative values is said to have an exponential distribution with parameter $\theta > 0$, if its probability density function is given by

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

Problem.1

Find the moment generating function of Exponential distribution and find its mean and variance.

Solution:

We know that $f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$

$$\begin{aligned} M_X(t) &= E(e^{tx}) = \int_0^{\infty} e^{tx} f(x) dx = \int_0^{\infty} \lambda e^{-\lambda x} e^{tx} dx \\ &= \lambda \int_0^{\infty} e^{-x(\lambda-t)} dx \\ &= \lambda \left[\frac{e^{-x(\lambda-t)}}{-(\lambda-t)} \right]_0^{\infty} = \frac{\lambda}{\lambda-t} \end{aligned}$$

$$\begin{aligned} \text{Mean} = \mu_1' &= \left[\frac{d}{dt} M_X(t) \right]_{t=0} = \left[\frac{\lambda}{(\lambda-t)^2} \right]_{t=0} = \frac{1}{\lambda} \\ \mu_2' &= \left[\frac{d^2}{dt^2} M_X(t) \right]_{t=0} = \left[\frac{\lambda(2)}{(\lambda-t)^3} \right]_{t=0} = \frac{2}{\lambda^2} \end{aligned}$$

$$\text{Variance} = \mu_2' - (\mu_1')^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}.$$

Problem.2

State and prove the memoryless property of exponential distribution.

Solution:

Statement:

If X is exponentially distributed with parameters λ , then for any two positive integers 's' and 't', $P[X > s+t / X > s] = P[X > t]$

Proof:

The p.d.f of X is $f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{Otherwise} \end{cases}$

$$\therefore P[X > k] = \int_k^{\infty} \lambda e^{-\lambda x} dx = \left[-e^{-\lambda x} \right]_k^{\infty} = e^{-\lambda k}$$

$$\begin{aligned} \therefore P[X > s+t / X > s] &= \frac{P[X > s+t \cap X > s]}{P[X > s]} \\ &= \frac{P[X > s+t]}{P[X > s]} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} \\ &= P[X > t] \end{aligned}$$

Problem.3

A component has an exponential time to failure distribution with mean of 10,000 hours.

(i) The component has already been in operation for its mean life. What is the probability that it will fail by 15,000 hours?

(ii) At 15,000 hours the component is still in operation. What is the probability that it will operate for another 5000 hours.

Solution:

Let X denote the time to failure of the component then X has exponential distribution with $Mean = 1000$ hours.

$$\therefore \frac{1}{\lambda} = 10,000 \Rightarrow \lambda = \frac{1}{10,000}$$

$$\text{The p.d.f. of } X \text{ is } f(x) = \begin{cases} \frac{1}{10,000} e^{-\frac{x}{10,000}}, & x \geq 0 \\ 0 & , otherwise \end{cases}$$

(i) Probability that the component will fail by 15,000 hours given it has already been in operation for its mean life = $P[x < 15,000 / x > 10,000]$

$$\begin{aligned} &= \frac{P[10,000 < X < 15,000]}{P[X > 10,000]} \\ &= \frac{\int_{10,000}^{15,000} f(x) dx}{\int_{10,000}^{\infty} f(x) dx} = \frac{e^{-1} - e^{-1.5}}{e^{-1}} \\ &= \frac{0.3679 - 0.2231}{0.3679} = 0.3936. \end{aligned}$$

(ii) Probability that the component will operate for another 5000 hours given that it is in operational 15,000 hours = $P[X > 20,000 / X > 15,000]$

$$\begin{aligned} &= P[x > 5000] \quad [\text{By memoryless prop}] \\ &= \int_{5000}^{\infty} f(x) dx = e^{-0.5} = 0.6065 \end{aligned}$$

Normal distribution:

A random variable X is said to have a Normal distribution with parameters μ (mean) and σ^2 (variance) if its probability density function is given by the probability law

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, -\infty < x < \infty, -\infty < \mu < \infty, \sigma > 0$$

Notation: $X \sim N(\mu, \sigma^2)$ read as X is following normal distribution with mean μ and variance σ^2 are called parameter.

Problem.1

Prove that "For standard normal distribution $N(0,1)$, $M_X(t) = e^{\frac{t^2}{2}}$."

Solution:

Moment generating function of Normal distribution

$$= M_X(t) = E[e^{tx}]$$

$$= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

Put $z = \frac{x-\mu}{\sigma}$ then $\sigma dz = dx$, $-\infty < Z < \infty$

$$\begin{aligned} \therefore M_X(t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{t(\sigma z + \mu) - \frac{z^2}{2}} dz \\ &= \frac{e^{\mu t}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\left(\frac{z^2}{2} - t\sigma z\right)} dz \\ &= \frac{e^{\mu t}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z-t\sigma)^2 + \left(\frac{\sigma^2 t^2}{2}\right)} dz \\ &= \frac{e^{\mu t} e^{\frac{\sigma^2 t^2}{2}}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z-t\sigma)^2} dz \end{aligned}$$

$\therefore$ the total area under normal curve is unity, we have $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z-t\sigma)^2} dz = 1$

Hence $M_X(t) = e^{\mu t + \frac{\sigma^2 t^2}{2}}$. $\therefore$ For standard normal variable $N(0,1)$

$$M_X(t) = e^{\frac{t^2}{2}}$$

Problem.2

State and prove the additive property of normal distribution.

Solution:

Statement:

If $X_1, X_2, \dots, X_n$ are n independent normal random variates with mean $(\mu_1, \sigma_1^2), (\mu_2, \sigma_2^2), \dots, (\mu_n, \sigma_n^2)$ then $X_1 + X_2 + \dots + X_n$ also a normal random variable with mean $\left(\sum_{i=1}^n \mu_i, \sum_{i=1}^n \sigma_i^2\right)$.

Proof:

We know that. $M_{X_1+X_2+\dots+X_n}(t) = M_{X_1}(t)M_{X_2}(t)\dots M_{X_n}(t)$

But $M_{X_i}(t) = e^{\mu_i t + \frac{t^2 \sigma_i^2}{2}}$, $i = 1, 2, \dots, n$

$$\begin{aligned} M_{X_1+X_2+\dots+X_n}(t) &= e^{\mu_1 t + \frac{t^2 \sigma_1^2}{2}} e^{\mu_2 t + \frac{t^2 \sigma_2^2}{2}} \dots e^{\mu_n t + \frac{t^2 \sigma_n^2}{2}} \\ &= e^{(\mu_1 + \mu_2 + \dots + \mu_n)t + \frac{(\sigma_1^2 + \sigma_2^2 + \dots + \sigma_n^2)t^2}{2}} \\ &= e^{\sum_{i=1}^n \mu_i t + \frac{\sum_{i=1}^n \sigma_i^2 t^2}{2}} \end{aligned}$$

By uniqueness MGF, $X_1 + X_2 + \dots + X_n$ follows normal random variable with parameter $\left(\sum_{i=1}^n \mu_i, \sum_{i=1}^n \sigma_i^2\right)$.

This proves the property.

Problem.3

X is a normal variate with $mean = 30$ and $S.D = 5$ Find the following $P[26 \leq X \leq 40]$

Solution:

$$X \sim N(30, 5^2)$$

$$\therefore \mu = 30 \text{ \& } \sigma = 5$$

Let $Z = \frac{X - \mu}{\sigma}$ be the standard normal variate

$$\begin{aligned} P[26 \leq X \leq 40] &= P\left[\frac{26-30}{5} \leq Z \leq \frac{40-30}{5}\right] \\ &= P[-0.8 \leq Z \leq 2] = P[-0.8 \leq Z \leq 0] + P[0 \leq Z \leq 2] \\ &= P[0 \leq Z \leq 0.8] + [0 \leq z \leq 2] \\ &= 0.2881 + 0.4772 = 0.7653. \end{aligned}$$

Problem.4

The average percentage of marks of candidates in an examination is 45 with a standard deviation of 10 the minimum for a pass is 50%. If 1000 candidates appear for the examination, how many can be expected marks. If it is required, that double that number should pass, what should be the average percentage of marks?

Solution:

Let X be marks of the candidates

$$\text{Then } X \sim N(42, 10^2)$$

$$\text{Let } z = \frac{X - 42}{10}$$

$$P[X > 50] = P[Z > 0.8]$$

$$= 0.5 - P[0 < z < 0.8]$$

$$= 0.5 - 0.2881 = 0.2119$$

Since 1000 students write the test, nearly 212 students would pass the examination.

If double that number should pass, then the no of passes should be 424.

We have to find z_1 , such that $P[Z > z_1] = 0.424$

$$\therefore P[0 < z < z_1] = 0.5 - 0.424 = 0.076$$

From tables, $z = 0.19$

$$\therefore z_1 = \frac{50 - x_1}{10} \Rightarrow x_1 = 50 - 10z_1$$

$$= 50 - 1.9 = 48.1$$

The average mark should be 48 nearly.

Problem.5

Given that X is normally distribution with mean 10 and probability $P[X > 12] = 0.1587$.

What is the probability that X will fall in the interval (9,11).

Solution:

Given X is normally distributed with mean $\mu = 10$.

Let $z = \frac{x - \mu}{\sigma}$ be the standard normal variate.

$$\text{For } X = 12, z = \frac{12 - 10}{\sigma} \Rightarrow z = \frac{2}{\sigma}$$

$$\text{Put } z_1 = \frac{2}{\sigma}$$

$$\text{Then } P[X > 12] = 0.1587$$

$$P[Z > z_1] = 0.1587$$

$$\therefore 0.5 - P[0 < z < z_1] = 0.1587$$

$$\Rightarrow P[0 < z < z_1] = 0.3413$$

From area table $P[0 < z < 1] = 0.3413$

$$\therefore z_1 = 1 \Rightarrow \frac{2}{\sigma} = 1$$

To find $P[9 < x < 11]$

$$\text{For } X = 9, z = -\frac{1}{2} \text{ and } X = 11, z = \frac{1}{2}$$

$$\therefore P[9 < X < 11] = P[-0.5 < z < 0.5]$$

$$= 2P[0 < z < 0.5]$$

$$= 2 \times 0.1915 = 0.3830$$

31. In a normal distribution 31% of the items are under 45 and 8% are over 64. Find the mean and standard deviation of the distribution.

Solution:

Let μ be the mean and σ be the standard deviation.

Then $P[X \leq 45] = 0.31$ and $P[X \geq 64] = 0.08$

$$\text{When } X = 45, Z = \frac{45 - \mu}{\sigma} = -z_1$$

$$\therefore z_1 \text{ is the value of } z \text{ corresponding to the area } \int_0^{z_1} \phi(z) dz = 0.19$$

$$\therefore z_1 = 0.495$$

$$45 - \mu = -0.495\sigma \text{ ---(1)}$$

$$\text{When } X = 64, Z = \frac{64 - \mu}{\sigma} = z_2$$

$$\therefore z_2 \text{ is the value of } z \text{ corresponding to the area } \int_0^{z_2} \phi(z) dz = 0.42$$

$$\therefore z_2 = 1.405$$

$$64 - \mu = 1.405\sigma \text{ ---(2)}$$

Solving (1) & (2) We get $\mu = 10$ (approx) & $\sigma = 50$ (approx)

Probability and Statistics

TWO DIMENSIONAL RANDOM VARIABLE

1. Let X and Y have joint density function $f(x, y) = 2, 0 < x < y < 1$. Find the marginal density function. Find the conditional density function Y given $X = x$.

Solution:

Marginal density function of X is given by

$$\begin{aligned} f_X(x) &= f(x) = \int_{-\infty}^{\infty} f(x, y) dy \\ &= \int_x^1 f(x, y) dy = \int_x^1 2 dy = 2(y)_x^1 \\ &= 2(1-x), 0 < x < 1. \end{aligned}$$

Marginal density function of Y is given by

$$\begin{aligned} f_Y(y) &= f(y) = \int_{-\infty}^{\infty} f(x, y) dx \\ &= \int_0^y 2 dx = 2y, 0 < y < 1. \end{aligned}$$

Conditional distribution function of Y given $X = x$ is $f\left(\frac{y}{x}\right) = \frac{f(x, y)}{f(x)} = \frac{2}{2(1-x)} = \frac{1}{1-x}$.

2. Verify that the following is a distribution function. $F(x) = \begin{cases} 0 & , x < -a \\ \frac{1}{2} \left(\frac{x}{a} + 1 \right) & , -a < x < a \\ 1 & , x > a \end{cases}$

Solution:

$F(x)$ is a distribution function only if $f(x)$ is a density function.

$$f(x) = \frac{d}{dx} [F(x)] = \frac{1}{2a}, \quad -a < x < a$$

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\begin{aligned} \therefore \int_{-a}^a \frac{1}{2a} dx &= \frac{1}{2a} [x]_{-a}^a = \frac{1}{2a} [a - (-a)] \\ &= \frac{1}{2a} \cdot 2a = 1. \end{aligned}$$

Therefore, it is a distribution function.

3. Prove that $\int_{x_1}^{x_2} f_X(x) dx = P(x_1 < x < x_2)$

Solution:

$$\begin{aligned}\int_{x_1}^{x_2} f_X(x) dx &= [F_X(x)]_{x_1}^{x_2} \\ &= F_X(x_2) - F_X(x_1) \\ &= P[X \leq x_2] - P[X \leq x_1] \\ &= P[x_1 \leq X \leq x_2]\end{aligned}$$

4. A continuous random variable X has a probability density function $f(x) = 3x^2$, $0 \leq x \leq 1$.

Find 'a' such that $P(X \leq a) = P(X > a)$.

Solution:

Since $P(X \leq a) = P(X > a)$, each must be equal to $\frac{1}{2}$ because the probability is always 1.

$$\therefore P(X \leq a) = \frac{1}{2}$$

$$\Rightarrow \int_0^a f(x) dx = \frac{1}{2}$$

$$\int_0^a 3x^2 dx = \frac{1}{2} \Rightarrow 3 \left[\frac{x^3}{3} \right]_0^a = a^3 = \frac{1}{2}.$$

$$\therefore a = \left(\frac{1}{2} \right)^{\frac{1}{3}}$$

5. Suppose that the joint density function $f(x, y) = \begin{cases} Ae^{-x-y}, & 0 \leq x \leq y, \quad 0 \leq y \leq \infty \\ 0, & \text{otherwise} \end{cases}$ Determine A .

Solution:

Since $f(x, y)$ is a joint density function

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1.$$

$$\Rightarrow \int_0^{\infty} \int_0^y Ae^{-x} e^{-y} dx dy = 1$$

$$\Rightarrow A \int_0^{\infty} e^{-y} \left(\frac{e^{-x}}{-1} \right)_0^y dy = 1$$

$$\Rightarrow A \int_0^{\infty} [e^{-y} - e^{-2y}] dy = 1$$

$$\Rightarrow A \left[\frac{e^{-y}}{-1} - \frac{e^{-2y}}{-2} \right]_0^{\infty} = 1$$

$$\Rightarrow A \left[\frac{1}{2} \right] = 1 \Rightarrow A = 2$$

6. Examine whether the variables X and Y are independent, whose joint density function is $f(x, y) = xe^{-x(y+1)}$, $0 < x, y < \infty$.

Solution:

The marginal probability function of X is

$$f_X(x) = f(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_0^{\infty} xe^{-x(y+1)} dy$$

$$= x \left[\frac{e^{-x(y+1)}}{-x} \right]_0^{\infty} = -[0 - e^{-x}] = e^{-x},$$

The marginal probability function of Y is

$$f_Y(y) = f(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_0^{\infty} xe^{-x(y+1)} dx$$

$$= x \left\{ \left[\frac{e^{-x(y+1)}}{-(y+1)} \right]_0^{\infty} - \left[\frac{e^{-x(y+1)}}{(y+1)^2} \right]_0^{\infty} \right\}$$

$$= \frac{1}{(y+1)^2}$$

$$\text{Here } f(x) \cdot f(y) = e^{-x} \times \frac{1}{(1+y)^2} \neq f(x, y)$$

$\therefore X$ and Y are not independent.

7. If X has an exponential distribution with parameter 1. Find the pdf of $y = \sqrt{x}$

Solution:

$$\text{Since } y = \sqrt{x}, x = y^2$$

Since X has an exponential distribution with parameter 1, the pdf of X is given by $f_X(x) = e^{-x}, x > 0$ $f(x) = \lambda e^{-\lambda x}, \lambda = 1$

$$\therefore f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

$$f_Y(y) = e^{-x} 2y = 2ye^{-y^2}, y > 0$$

8. If X is uniformly distributed random variable in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, Find the probability density function of $Y = \tan X$.

Solution:

Given $Y = \tan X \Rightarrow x = \tan^{-1} y$

$$\therefore \frac{dx}{dy} = \frac{1}{1+y^2}$$

Since X is uniformly distribution in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$,

$$f_X(x) = \frac{1}{b-a} = \frac{1}{\frac{\pi}{2} - \left(-\frac{\pi}{2}\right)}$$

$$f_X(x) = \frac{1}{\pi}, -\frac{\pi}{2} < x < \frac{\pi}{2}$$

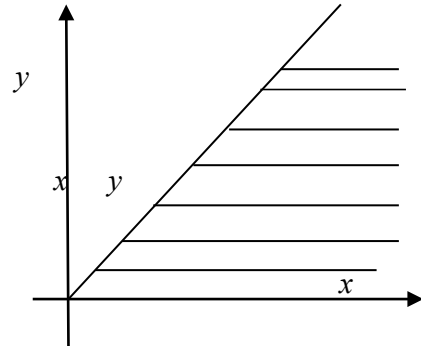
$$\text{Now } f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right| = \frac{1}{\pi} \left(\frac{1}{1+y^2} \right), -\infty < y < \infty$$

$$\therefore f_Y(y) = \frac{1}{\pi(1+y^2)}, -\infty < y < \infty$$

9. If the Joint probability density function of (x, y) is given by $f(x, y) = 24y(1-x)$, $0 \leq y \leq x \leq 1$ Find $E(XY)$.

Solution:

$$\begin{aligned} E(xy) &= \int_0^1 \int_y^1 xyf(x, y) dx dy \\ &= 24 \int_0^1 \int_y^1 xy^2(1-x) dx dy \\ &= 24 \int_0^1 y^2 \left[\frac{1}{6} - \frac{y^2}{2} + \frac{y^3}{3} \right] dy = \frac{4}{15} \end{aligned}$$



10. If X and Y are random Variables, Prove that $Cov(X, Y) = E(XY) - E(X)E(Y)$

Solution:

$$\begin{aligned} cov(X, Y) &= E[(X - E(X))(Y - E(Y))] \\ &= E(XY - \bar{X}Y - \bar{Y}X + \bar{X}\bar{Y}) \\ &= E(XY) - \bar{X}E(Y) - \bar{Y}E(X) + \bar{X}\bar{Y} \\ &= E(XY) - \bar{X}\bar{Y} - \bar{X}\bar{Y} + \bar{X}\bar{Y} \\ &= E(XY) - E(X)E(Y) \quad \left[E(X) = \bar{X}, E(Y) = \bar{Y} \right] \end{aligned}$$

11. If X and Y are independent random variables prove that $cov(x, y) = 0$

Proof:

$$\text{cov}(x, y) = E(xy) - E(x)E(y)$$

But if X and Y are independent then $E(xy) = E(x)E(y)$

$$\text{cov}(x, y) = E(x)E(y) - E(x)E(y)$$

$$\text{cov}(x, y) = 0.$$

12. Write any two properties of regression coefficients.

Solution:

1. Correction coefficients is the geometric mean of regression coefficients
2. If one of the regression coefficients is greater than unity then the other should be less than 1.

$$b_{xy} = r \frac{\sigma_y}{\sigma_x} \text{ and } b_{yx} = r \frac{\sigma_x}{\sigma_y}$$

If $b_{xy} > 1$ then $b_{yx} < 1$.

13. Write the angle between the regression lines.

The slopes of the regression lines are

$$m_1 = r \frac{\sigma_y}{\sigma_x}, m_2 = \frac{1}{r} \frac{\sigma_x}{\sigma_y}$$

If θ is the angle between the lines, Then

$$\tan\theta = \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \left[\frac{1 - r^2}{r} \right]$$

When $r = 0$, that is when there is no correlation between x and y , $\tan\theta = \infty$ (or) $\theta = \frac{\pi}{2}$

and so the regression lines are perpendicular

When $r = 1$ or $r = -1$, that is when there is a perfect correlation +ve or -ve, $\theta = 0$ and so the lines coincide.

15. i). Two random variables are said to be orthogonal if correction is zero.
 ii). If $X = Y$ then correlation coefficients between them is 1.

16.a). The joint probability density function of a bivariate random variable X, Y is

$$f_{XY}(x, y) = \begin{cases} k(x+y), & 0 < x < 2, 0 < y < 2 \\ 0, & \text{otherwise} \end{cases} \quad \text{where 'k' is a constant.}$$

- Find k .
- Find the marginal density function of X and Y .
- Are X and Y independent?
Find $f_X(x)$ and $f_Y(y)$.
- Find $f_{X|Y}(x/y)$ and $f_{Y|X}(y/x)$.

Solution:

(i). Given the joint probability density function of a bivariate random variable (X, Y) is

$$f_{XY}(x, y) = \begin{cases} K(x+y), & 0 < x < 2, 0 < y < 2 \\ 0, & \text{otherwise} \end{cases}$$

$$\text{Here } \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y) dx dy = 1 \Rightarrow \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} K(x+y) dx dy = 1$$

$$\int_0^2 \int_0^2 K(x+y) dx dy = 1 \Rightarrow K \int_0^2 \left[\frac{x^2}{2} + xy \right]_0^2 dy = 1$$

$$\Rightarrow K \int_0^2 (2+2y) dy = 1$$

$$\Rightarrow K [2y + y^2]_0^2 = 1$$

$$\Rightarrow K [8 - 0] = 1$$

$$\Rightarrow K = \frac{1}{8}$$

(ii). The marginal p.d.f of X is given by

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f(x, y) dy = \frac{1}{8} \int_0^2 (x+y) dy \\ &= \frac{1}{8} \left[xy + \frac{y^2}{2} \right]_0^2 = \frac{1+x}{4} \end{aligned}$$

$\therefore$ The marginal p.d.f of X is

$$f_X(x) = \begin{cases} \frac{x+1}{4}, & 0 < x < 2 \\ 0, & \text{otherwise} \end{cases}$$

The marginal p.d.f of Y is

$$\begin{aligned} f_Y(y) &= \int_{-\infty}^{\infty} f(x, y) dx = \frac{1}{8} \int_0^2 (x+y) dx \\ &= \frac{1}{8} \left[\frac{x^2}{2} + yx \right]_0^2 \end{aligned}$$

$$= \frac{1}{8} [2 + 2y] = \frac{y+1}{4}$$

∴ The marginal p.d.f of Y is

$$f_Y(y) = \begin{cases} \frac{y+1}{4}, & 0 < y < 2 \\ 0, & \text{otherwise} \end{cases}$$

(iii). To check whether X and Y are independent or not.

$$f_X(x) f_Y(y) = \frac{(x+1)}{4} \frac{(y+1)}{4} \neq f_{XY}(x, y)$$

Hence X and Y are not independent.
(iv). Conditional p.d.f $f_{Y/X}(y/x)$ is given by

$$f_{Y/X}(y/x) = \frac{f(x, y)}{f_X(x)} = \frac{\frac{1}{8}(x+y)}{\frac{1}{4}(x+1)} = \frac{1}{2} \frac{(x+y)}{(x+1)}$$

$$f_{Y/X}(y/x) = \frac{1}{2} \left(\frac{x+y}{x+1} \right), \quad 0 < x < 2, \quad 0 < y < 2$$

$$\begin{aligned} \text{(v)} \quad P \left(0 < y < \frac{1}{2} \middle| x=1 \right) &= \int_0^{\frac{1}{2}} f_{Y/X}(y/x=1) dy \\ &= \frac{1}{2} \int_0^{\frac{1}{2}} \frac{1+y}{2} dy = \frac{5}{32}. \end{aligned}$$

17.a). If X and Y are two random variables having joint probability density function

$$f(x, y) = \begin{cases} \frac{1}{8}(6-x-y), & 0 < x < 2, \quad 2 < y < 4 \\ 0, & \text{otherwise} \end{cases} \quad \text{Find (i) } P(X < 1 \cap Y < 3)$$

(ii) $P(X+Y < 3)$ (iii) $P(X < 1/Y < 3)$.

b). Three balls are drawn at random without replacement from a box containing 2 white, 3 red and 4 black balls. If X denotes the number of white balls drawn and Y denotes the number of red balls drawn find the joint probability distribution of X, Y .

Solution:

a).

$$\begin{aligned} P(X < 1 \cap Y < 3) &= \int_{y=-\infty}^{y=3} \int_{x=-\infty}^{x=1} f(x, y) dx dy \\ &= \int_{y=2}^{y=3} \int_{x=0}^{x=1} \frac{1}{8}(6-x-y) dx dy \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{8} \int_2^3 \int_0^1 (6-x-y) dx dy \\
&= \frac{1}{8} \int_2^3 \left[6x - \frac{x^2}{2} - xy \right]_0^1 dy \\
&= \frac{1}{8} \int_2^3 \left[\frac{11}{2} - y \right] dy = \frac{1}{8} \left[\frac{11y}{2} - \frac{y^2}{2} \right]_2^3
\end{aligned}$$

$$P(X < 1 \cap Y < 3) = \frac{3}{8}$$

$$\begin{aligned}
\text{(ii). } P(X+Y < 3) &= \int_0^1 \int_2^{3-x} \frac{1}{8} (6-x-y) dy dx \\
&= \frac{1}{8} \int_0^1 \left[6y - xy - \frac{y^2}{2} \right]_2^{3-x} dx \\
&= \frac{1}{8} \int_0^1 \left[6(3-x) - x(3-x) - \frac{(3-x)^2}{2} - [12 - 2x - 2] \right] dx \\
&= \frac{1}{8} \int_0^1 \left[18 - 6x - 3x + x^2 - \frac{(9+x^2-6x)}{2} - (10-2x) \right] dx \\
&= \frac{1}{8} \int_0^1 \left[18 - 9x + x^2 - \frac{9}{2} - \frac{x^2}{2} + \frac{6x}{2} - 10 + 2x \right] dx \\
&= \frac{1}{8} \int_0^1 \left[\frac{7}{2} - 4x + \frac{x^2}{2} \right] dx \\
&= \frac{1}{8} \left[\frac{7x}{2} - \frac{4x^2}{2} + \frac{x^3}{6} \right]_0^1 = \frac{1}{8} \left[\frac{7}{2} - 2 + \frac{1}{6} \right] \\
&= \frac{1}{8} \left[\frac{21-12+1}{6} \right] = \frac{1}{8} \left(\frac{10}{6} \right) = \frac{5}{24}.
\end{aligned}$$

$$\text{(iii). } P(X < 1 / Y < 3) = \frac{P(x < 1 \cap y < 3)}{P(y < 3)}$$

$$\begin{aligned}
\text{The Marginal density function of Y is } f_Y(y) &= \int_0^2 f(x, y) dx \\
&= \int_0^2 \frac{1}{8} (6-x-y) dx \\
&= \frac{1}{8} \left[6x - \frac{x^2}{2} - yx \right]_0^2 \\
&= \frac{1}{8} [12 - 2 - 2y]
\end{aligned}$$

$$= \frac{5-y}{4}, 2 < y < 4.$$

$$\begin{aligned} P(X < 1/Y < 3) &= \frac{\int_{x=0}^{x=1} \int_{y=2}^{y=3} \frac{1}{8} (6-x-y) dx dy}{\int_{y=2}^{y=3} f_Y(y) dy} \\ &= \frac{\frac{3}{8}}{\int_2^3 \left(\frac{5-y}{4} \right) dy} = \frac{\frac{3}{8}}{\frac{1}{4} \left[5y - \frac{y^2}{2} \right]_2^3} \\ &= \frac{3}{8} \times \frac{8}{5} = \frac{3}{5}. \end{aligned}$$

b). Let X takes 0, 1, 2 and Y takes 0, 1, 2 and 3.

$P(X=0, Y=0) = P(\text{drawing 3 balls none of which is white or red})$

$P(\text{all the 3 balls drawn are black})$

$$= \frac{4C_3}{9C_3} = \frac{4 \times 3 \times 2 \times 1}{9 \times 8 \times 7} = \frac{1}{21}.$$

$P(X=0, Y=1) = P(\text{drawing 1 red ball and 2 black balls})$

$$= \frac{3C_1 \times 4C_2}{9C_3} = \frac{3}{14}$$

$P(X=0, Y=2) = P(\text{drawing 2 red balls and 1 black ball})$

$$= \frac{3C_2 \times 4C_1}{9C_3} = \frac{3 \times 2 \times 4 \times 3}{9 \times 8 \times 7} = \frac{1}{7}.$$

$P(X=0, Y=3) = P(\text{all the three balls drawn are red and no white ball})$

$$= \frac{3C_3}{9C_3} = \frac{1}{84}$$

$P(X=1, Y=0) = P(\text{drawing 1 White and no red ball})$

$$\begin{aligned} &= \frac{2C_1 \times 4C_2}{9C_3} = \frac{\frac{2 \times 4 \times 3}{1 \times 2}}{\frac{9 \times 8 \times 7}{1 \times 2 \times 3}} \\ &= \frac{12 \times 1 \times 2 \times 3}{9 \times 8 \times 7} = \frac{1}{7}. \end{aligned}$$

$P(X=1, Y=1) = P(\text{drawing 1 White and 1 red ball})$

$$= \frac{2C_1 \times 3C_1}{9C_3} = \frac{\frac{2 \times 3}{9 \times 8 \times 7}}{\frac{1 \times 2 \times 3}{1 \times 2 \times 3}} = \frac{2}{7}$$

$$P(X = 1, Y = 2) = P(\text{drawing 1 White and 2 red ball})$$

$$= \frac{{}^2C_1 \times {}^3C_2}{{}^9C_3} = \frac{2 \times 3 \times 2}{9 \times 8 \times 7} = \frac{1}{14}$$

$$P(X = 1, Y = 3) = 0 \text{ (Since only three balls are drawn)}$$

$$P(X = 2, Y = 0) = P(\text{drawing 2 white balls and no red balls})$$

$$= \frac{{}^2C_2 \times {}^4C_1}{{}^9C_3} = \frac{1}{21}$$

$$P(X = 2, Y = 1) = P(\text{drawing 2 white balls and no red balls})$$

$$= \frac{{}^2C_2 \times {}^3C_1}{{}^9C_3} = \frac{1}{28}$$

$$P(X = 2, Y = 2) = 0$$

$$P(X = 2, Y = 3) = 0$$

The joint probability distribution of X, Y may be represented as

$X \backslash Y$	0	1	2	3
0	$\frac{1}{21}$	$\frac{3}{14}$	$\frac{1}{7}$	$\frac{1}{84}$
1	$\frac{1}{7}$	$\frac{2}{7}$	$\frac{1}{14}$	0
2	$\frac{1}{21}$	$\frac{1}{28}$	0	0

18.a). Two fair dice are tossed simultaneously. Let X denotes the number on the first die and Y denotes the number on the second die. Find the following probabilities.

$$(i) \ P(X + Y) = 8, \ (ii) \ P(X + Y \geq 8), \ (iii) \ P(X = Y) \text{ and } (iv) \ P(X + Y = 6 \bigg/ Y = 4).$$

b) The joint probability mass function of a bivariate discrete random variable (X, Y) in given by the table.

$Y \backslash X$	1	2	3
1	0.1	0.1	0.2
2	0.2	0.3	0.1

Find

- The marginal probability mass function of X and Y .
- The conditional distribution of X given $Y = 1$.
- $P(X + Y < 4)$

Solution:

a). Two fair dice are thrown simultaneously

$$S = \left\{ \begin{array}{l} (1,1)(1,2)\dots(1,6) \\ (2,1)(2,2)\dots(2,6) \\ \vdots \quad \vdots \quad \dots \quad \vdots \\ (6,1)(6,2)\dots(6,6) \end{array} \right\}, \quad n(S) = 36$$

Let X denotes the number on the first die and Y denotes the number on the second die.

Joint probability density function of (X, Y) is $P(X = x, Y = y) = \frac{1}{36}$ for

$$x = 1, 2, 3, 4, 5, 6 \text{ and } y = 1, 2, 3, 4, 5, 6$$

(i) $X + Y = \{ \text{the events that the no is equal to 8} \}$

$$= \{(2,6), (3,5), (4,4), (5,3), (6,2)\}$$

$$P(X + Y = 8) = P(X = 2, Y = 6) + P(X = 3, Y = 5) + P(X = 4, Y = 4)$$

$$+ P(X = 5, Y = 3) + P(X = 6, Y = 2)$$

$$= \frac{1}{36} + \frac{1}{36} + \frac{1}{36} + \frac{1}{36} + \frac{1}{36} = \frac{5}{36}$$

(ii) $P(X + Y \geq 8)$

$$X + Y = \left\{ \begin{array}{l} (2,6) \\ (3,5), (3,6) \\ (4,4), (4,5), (4,6) \\ (5,3), (5,4), (5,5), (5,6) \\ (6,2), (6,3), (6,4), (6,5), (6,6) \end{array} \right\}$$

$$\therefore P(X + Y \geq 8) = P(X + Y = 8) + P(X + Y = 9) + P(X + Y = 10)$$

$$+ P(X + Y = 11) + P(X + Y = 12)$$

$$= \frac{5}{36} + \frac{4}{36} + \frac{3}{36} + \frac{2}{36} + \frac{1}{36} = \frac{15}{36} = \frac{5}{12}$$

(iii) $P(X = Y)$

$$P(X = Y) = P(X = 1, Y = 1) + P(X = 2, Y = 2) + \dots + P(X = 6, Y = 6)$$

$$= \frac{1}{36} + \frac{1}{36} + \dots + \frac{1}{36} = \frac{6}{36} = \frac{1}{6}$$

$$(iv) P(X + Y = 6 / Y = 4) = \frac{P(X + Y = 6 \cap Y = 4)}{P(Y = 4)}$$

$$\text{Now } P(X + Y = 6 \cap Y = 4) = \frac{1}{36}$$

$$P(Y = 4) = \frac{6}{36}$$

$$\therefore P(X+Y=6/Y=4)=\frac{\frac{1}{36}}{\frac{1}{6}}=\frac{1}{6}.$$

b). The joint probability mass function of (X,Y) is

$\begin{matrix} X \\ Y \end{matrix}$	1	2	3	Total
1	0.1	0.1	0.2	0.4
2	0.2	0.3	0.1	0.6
Total	0.3	0.4	0.3	1

From the definition of marginal probability function

$$P_X(x_i)=\sum_{y_j}P_{XY}(x_i,y_j)$$

When $X=1$,

$$\begin{aligned} P_X(x_i) &= P_{XY}(1,1)+P_{XY}(1,2) \\ &= 0.1+0.2=0.3 \end{aligned}$$

When $X=2$,

$$\begin{aligned} P_X(x=2) &= P_{XY}(2,1)+P_{XY}(2,2) \\ &= 0.2+0.3=0.4 \end{aligned}$$

When $X=3$,

$$\begin{aligned} P_X(x=3) &= P_{XY}(3,1)+P_{XY}(3,2) \\ &= 0.2+0.1=0.3 \end{aligned}$$

$\therefore$ The marginal probability mass function of X is

$$P_X(x)=\begin{cases} 0.3 & \text{when } x=1 \\ 0.4 & \text{when } x=2 \\ 0 & \\ 0.3 & \text{when } x=3 \end{cases}$$

The marginal probability mass function of Y is given by $P_Y(y_j)=\sum_{x_i}P_{XY}(x_i,y_j)$

$$\begin{aligned} \text{When } Y=1, P_Y(y=1) &= \sum_{x_i=1}^3 P_{XY}(x_i,1) \\ &= P_{XY}(1,1)+P_{XY}(2,1)+P_{XY}(3,1) \\ &= 0.1+0.1+0.2=0.4 \end{aligned}$$

$$\begin{aligned} \text{When } Y=2, P_Y(y=2) &= \sum_{x_i=1}^3 P_{XY}(x_i,2) \\ &= P_{XY}(1,2)+P_{XY}(2,2)+P_{XY}(3,2) \\ &= 0.2+0.3+0.1=0.6 \end{aligned}$$

$\therefore$ Marginal probability mass function of Y is

$$P_Y(y)=\begin{cases} 0.4 & \text{when } y=1 \\ 0.6 & \text{when } y=2 \end{cases}$$

(ii) The conditional distribution of X given $Y=1$ is given by

$$P(X=x/Y=1)=\frac{P(X=x\cap Y=1)}{P(Y=1)}$$

From the probability mass function of Y , $P_{y=1}=P_y=0.4$

$$\begin{aligned}\text{When } X=1, P(X=1/Y=1) &= \frac{P(X=1\cap Y=1)}{P(Y=1)} \\ &= \frac{P_{XY}(1,1)}{P_Y(1)} = \frac{0.1}{0.4} = 0.25\end{aligned}$$

$$\text{When } X=2, P(X=2/Y=1)=\frac{P_{XY}(2,1)}{P_Y(1)}=\frac{0.1}{0.4}=0.25$$

$$\text{When } X=3, P(X=3/Y=1)=\frac{P_{XY}(3,1)}{P_Y(1)}=\frac{0.2}{0.4}=0.5$$

$$\begin{aligned}\text{(iii). } P(X+Y < 4) &= P\{(x,y)/x+y < 4 \text{ Where } x=1,2,3; y=1,2\} \\ &= P\{(1,1),(1,2),(2,1)\} \\ &= P_{XY}(1,1)+P_{XY}(1,2)+P_{XY}(2,1) \\ &= 0.1+0.1+0.2 = 0.4\end{aligned}$$

19.a). If X and Y are two random variables having the joint density function $f(x,y)=\frac{1}{27}(x+2y)$ where x and y can assume only integer values 0, 1 and 2, find the conditional distribution of Y for $X=x$.

b). The joint probability density function of X,Y is given by

$$f_{XY}(x,y)=\begin{cases} xy^2+\frac{x^2}{8}, & 0\leq x\leq 2, \quad 0\leq y\leq 1 \\ 0, & \text{otherwise} \end{cases}.$$

Find (i) $P(X=1)$, (ii) $P(X=Y)$ and

(iii) $P(X+Y=1)$

Solution:

a). Given X and Y are two random variables having the joint density function

$$f(x,y)=\frac{1}{27}(x+2y)-----(1)$$

Where $x=0,1,2$ and $y=0,1,2$

Then the joint probability distribution X and Y becomes as follows

$\begin{matrix} Y \\ X \end{matrix}$	0	1	2	$f_1(x)$
0	0	$\frac{1}{27}$	$\frac{2}{27}$	$\frac{3}{27}$

1	$\frac{2}{27}$	$\frac{3}{27}$	$\frac{4}{27}$	$\frac{9}{27}$
2	$\frac{4}{27}$	$\frac{5}{27}$	$\frac{6}{27}$	$\frac{15}{27}$

The marginal probability distribution of X is given by $f_1(X)=\sum_j P(x,y)$ and is calculated in

the above column of above table.
The conditional distribution of Y for X is given by $f_1\left(Y=y\bigg/X=x\right)=\frac{f(x,y)}{f_1(x)}$ and is obtained in

the following table.

$\begin{matrix} X \\ Y \end{matrix}$	0	1	2
0	0	$\frac{1}{3}$	$\frac{2}{3}$
1	$\frac{1}{9}$	$\frac{3}{9}$	$\frac{5}{9}$
2	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{2}$

$$P\left(Y=0\bigg/X=0\right)=\frac{P(X=0,Y=0)}{P(X=0)}=\frac{0}{\frac{6}{27}}=0$$

$$P\left(Y=1\bigg/X=0\right)=\frac{P(X=0,Y=1)}{P(X=0)}=\frac{\frac{2}{27}}{\frac{6}{27}}=\frac{1}{3}$$

$$P\left(Y=2\bigg/X=0\right)=\frac{P(X=0,Y=2)}{P(X=0)}=\frac{\frac{4}{27}}{\frac{6}{27}}=\frac{2}{3}$$

$$P\left(Y=0\bigg/X=1\right)=\frac{P(X=1,Y=0)}{P(X=1)}=\frac{\frac{1}{27}}{\frac{9}{27}}=\frac{1}{9}$$

$$P\left(Y=1\bigg/X=1\right)=\frac{P(X=1,Y=1)}{P(X=1)}=\frac{\frac{3}{27}}{\frac{9}{27}}=\frac{3}{9}=\frac{1}{3}$$

$$P\left(Y=2\bigg/X=1\right)=\frac{P(X=1,Y=2)}{P(X=1)}=\frac{\frac{5}{27}}{\frac{9}{27}}=\frac{5}{9}$$

$$P(Y=0/X=2) = \frac{P(X=2, Y=0)}{P(X=2)} = \frac{\frac{2}{27}}{\frac{12}{27}} = \frac{1}{6}$$

$$P(Y=1/X=2) = \frac{P(X=2, Y=1)}{P(X=2)} = \frac{\frac{4}{27}}{\frac{12}{27}} = \frac{1}{3}$$

$$P(Y=2/X=2) = \frac{P(X=2, Y=2)}{P(X=2)} = \frac{\frac{6}{27}}{\frac{12}{27}} = \frac{1}{2}$$

b). Given the joint probability density function of X, Y is $f_{XY}(x, y) = xy^2 + \frac{x^2}{8}$, $0 \leq x \leq 2, 0 \leq y \leq 1$

$$(i). P(X > 1) = \int_1^{\infty} f_X(x) dx$$

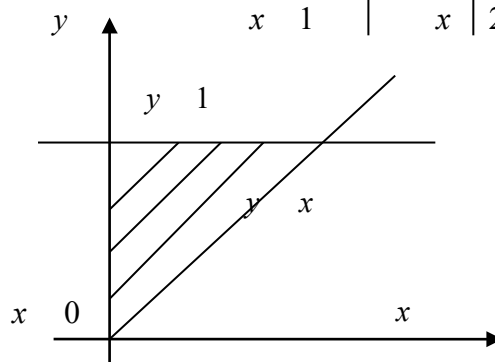
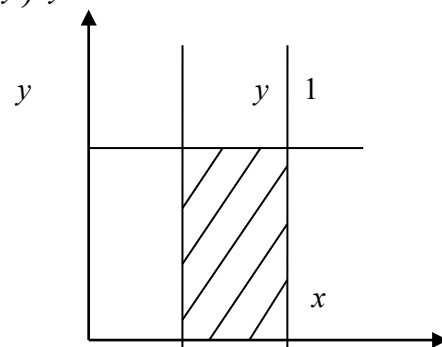
The Marginal density function of X is $f_X(x) = \int_0^1 f(x, y) dy$

$$\begin{aligned} f_X(x) &= \int_0^1 \left(xy^2 + \frac{x^2}{8} \right) dy \\ &= \left[\frac{xy^2}{3} + \frac{x^2 y}{8} \right]_0^1 = \frac{x}{3} + \frac{x^2}{8}, \quad 1 < x < 2 \end{aligned}$$

$$\begin{aligned} P(X > 1) &= \int_1^2 \left(\frac{x}{3} + \frac{x^2}{8} \right) dx \\ &= \left[\frac{x^2}{6} + \frac{x^3}{24} \right]_1^2 = \frac{19}{24}. \end{aligned}$$

$$(ii) P(X < Y) = \iint_{R_2} f_{XY}(x, y) dx dy$$

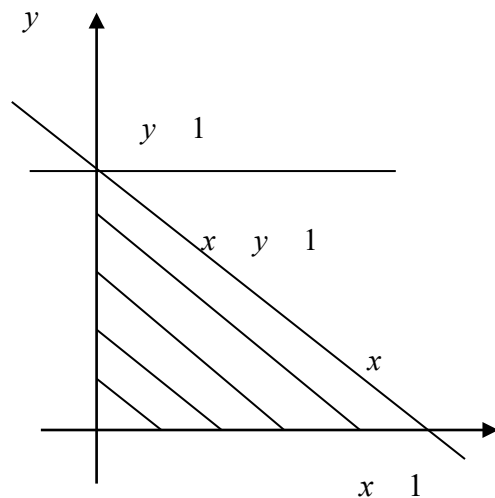
$$\begin{aligned} P(X < Y) &= \int_{y=0}^1 \int_{x=0}^y \left(xy^2 + \frac{x^2}{8} \right) dx dy \\ &= \int_0^1 \left[\frac{x^2 y^2}{2} + \frac{x^3}{24} \right]_0^y dy \\ &= \int_0^1 \left(\frac{y^4}{2} + \frac{y^3}{24} \right) dy = \left[\frac{y^5}{10} + \frac{y^4}{96} \right]_0^1 \\ &= \frac{1}{10} + \frac{1}{96} = \frac{96+10}{960} = \frac{53}{480} \end{aligned}$$



$$(iii) P(X+Y \leq 1) = \iint_{R_3} f_{XY}(x, y) dx dy$$

Where R_3 is the region

$$\begin{aligned} P(X+Y \leq 1) &= \int_{y=0}^1 \int_{x=0}^{1-y} \left(xy^2 + \frac{x^2}{8} \right) dx dy \\ &= \int_{y=0}^1 \left[\left(\frac{x^2 y^2}{2} + \frac{x^3}{24} \right) \right]_0^{1-y} dy \\ &= \int_{y=0}^1 \left(\frac{(1-y)^2 y^2}{2} + \frac{(1-y)^3}{24} \right) dy \\ &= \int_0^1 \left(\frac{(1+y^2-2y)y^2}{2} + \frac{(1-y)^3}{24} \right) dy \\ &= \left[\left[\frac{y^3}{3} + \frac{y^5}{5} - \frac{2y^2}{4} \right] \frac{1}{2} + \frac{(1-y)^4}{96} \right]_0^1 \\ &= \frac{1}{6} + \frac{1}{10} - \frac{1}{4} + \frac{1}{96} = \frac{13}{480}. \end{aligned}$$



20).a). If the joint distribution functions of X and Y is given by

$$F(x, y) = \begin{cases} (1-e^{-x})(1-e^{-y}), & x > 0, y > 0 \\ 0, & \text{otherwise} \end{cases}$$

- Find the marginal density of X and Y .
- Are X and Y independent.
- $P(1 < X < 3, 1 < Y < 2)$.

b). The joint probability distribution of X and Y is given by

$$f(x, y) = \begin{cases} \frac{6-x-y}{8}, & 0 < x < 2, 2 < y < 4 \\ 0, & \text{otherwise} \end{cases}. \text{ Find } P(1 < Y < 3 | X < 2).$$

Solution:

$$\begin{aligned} \text{a). Given } F(x, y) &= (1-e^{-x})(1-e^{-y}) \\ &= 1 - e^{-x} - e^{-y} + e^{-(x+y)} \end{aligned}$$

The joint probability density function is given by

$$\begin{aligned} f(x, y) &= \frac{\partial^2 F(x, y)}{\partial x \partial y} \\ &= \frac{\partial^2}{\partial x \partial y} [1 - e^{-x} - e^{-y} + e^{-(x+y)}] \end{aligned}$$

$$= \frac{\partial}{\partial x} [e^{-y} - e^{-(x+y)}]$$

$$\therefore f(x, y) = \begin{cases} e^{-(x+y)}, & x \geq 0, y \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

(ii) The marginal probability function of X is given by

$$f(x) = f_X(x)$$

$$= \int_{-\infty}^{\infty} f(x, y) dy = \int_0^{\infty} e^{-(x+y)} dy$$

$$= \left[\frac{e^{-(x+y)}}{-1} \right]_0^{\infty}$$

$$= \left[-e^{-(x+y)} \right]_0^{\infty}$$

$$= e^{-x}, \quad x > 0$$

The marginal probability function of Y is

$$f(y) = f_Y(y)$$

$$= \int_{-\infty}^{\infty} f(x, y) dx$$

$$= \int_0^{\infty} e^{-(x+y)} dx = \left[-e^{-(x+y)} \right]_0^{\infty}$$

$$= e^{-y}, \quad y > 0$$

$$\therefore f(x) f(y) = e^{-x} e^{-y} = e^{-(x+y)} = f(x, y)$$

$\therefore X$ and Y are independent.

(iii) $P(1 < X < 3, 1 < Y < 2) = P(1 < X < 3) \times P(1 < Y < 2)$ [Since X and Y are independent]

$$= \int_1^3 f(x) dx \times \int_1^2 f(y) dy.$$

$$= \int_1^3 e^{-x} dx \times \int_1^2 e^{-y} dy$$

$$= \left[\frac{e^{-x}}{-1} \right]_1^3 \times \left[\frac{e^{-y}}{-1} \right]_1^2$$

$$= (-e^{-3} + e^{-1})(-e^{-2} + e^{-1})$$

$$= e^{-5} - e^{-4} - e^{-3} + e^{-2}$$

b). $P(1 < Y < 3 | X > 2) = \int_1^3 f(y | x > 2) dy$

$$f_X(x) = \int_2^4 f(x, y) dy$$

$$\begin{aligned}
 &= \int_2^4 \left(\frac{6-x-y}{8} \right) dy \\
 &= \frac{1}{8} \left(6y - xy - \frac{y^2}{2} \right)_2^4 \\
 &= \frac{1}{8} (16 - 4x - 10 + 2x)
 \end{aligned}$$

$$f\left(\frac{y}{x}\right)=\frac{f(x,y)}{f(x)}=\frac{\frac{6-x-y}{8}}{\frac{6-2x}{8}}=\frac{6-x-y}{6-2x}$$

$$\begin{aligned}
 P(1 \leq Y \leq 3/X) &= \int_2^3 f\left(\frac{y}{x}\right) dy \\
 &= \int_2^3 \left(\frac{4-y}{2} \right) dy \\
 &= \frac{1}{2} \left[4y - \frac{y^2}{2} \right]_2^3 \\
 &= \frac{1}{2} \left[4y - \frac{y^2}{2} \right]_2^3 = \frac{1}{2} \left[14 - \frac{17}{2} \right] = \frac{11}{4}.
 \end{aligned}$$

Probability and Statistics Correlation and Regression

21).a). Two random variables X and Y have the following joint probability density

function $f(x, y) = \begin{cases} 2 - x - y, & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & , \text{otherwise} \end{cases}$. Find the marginal probability density function

of X and Y . Also find the covariance between X and Y .

b). If $f(x, y) = \frac{6 - x - y}{8}, 0 \leq x \leq 2, 2 \leq y \leq 4$ for a bivariate X, Y , find the correlation coefficient

Solution:

a) Given the joint probability density function $f(x, y) = \begin{cases} 2 - x - y, & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & , \text{otherwise} \end{cases}$

$$\begin{aligned} \text{Marginal density function of } X \text{ is } f_X(x) &= \int_{-\infty}^{\infty} f(x, y) dy \\ &= \int_0^1 (2 - x - y) dy \\ &= \left[2y - xy - \frac{y^2}{2} \right]_0^1 \\ &= 2 - x - \frac{1}{2} \end{aligned}$$

$$f_X(x) = \begin{cases} \frac{3}{2} - x, & 0 < x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Marginal density function of Y is $f_Y(y) = \int_0^1 (2 - x - y) dx$

$$= \left[2x - \frac{x^2}{2} - xy \right]_0^1$$

$$= \frac{3}{2} - y$$

$$f_Y(y) = \begin{cases} \frac{3}{2} - y, & 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Covariance of $(X, Y) = \text{Cov}(X, Y) = E(XY) - E(X)E(Y)$

$$E(X) = \int_0^1 x f_X(x) dx = \int_0^1 x \left(\frac{3}{2} - x \right) dx = \left[\frac{3}{2} \frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = \frac{5}{12}$$

$$E(Y) = \int_0^1 y f_Y(y) dy = \int_0^1 y \left(\frac{3}{2} - y \right) dy = \frac{5}{12}$$

$\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$

$$E(XY) = \int_0^1 \int_0^1 xy f(x, y) dx dy$$

$$= \int_0^1 \int_0^1 xy (2 - x - y) dx dy$$

$$= \int_0^1 \int_0^1 (2xy - x^2y - xy^2) dx dy$$

$$= \int_0^1 \left[\frac{2x^2y}{2} - \frac{x^3}{3}y - \frac{x^2}{2}y^2 \right]_0^1 dy$$

$$= \int_0^1 \left(y - \frac{1}{3} - \frac{y^2}{2} \right) dy$$

$$= \left[\frac{y^2}{2} - \frac{y}{3} - \frac{y^3}{6} \right]_0^1 = \frac{1}{6}$$

$$\text{Cov}(X, Y) = \frac{1}{6} - \frac{5}{12} \times \frac{5}{12}$$

$$= \frac{1}{6} - \frac{25}{144} = -\frac{1}{144}$$

b). Correlation coefficient $\rho_{XY} = \frac{E(XY) - E(X)E(Y)}{\sigma_X \sigma_Y}$

Marginal density function of X is

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_2^4 \left(\frac{6-x-y}{8} \right) dy = \frac{6-2x}{8}$$

Marginal density function of Y is

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_0^2 \left(\frac{6-x-y}{8} \right) dx = \frac{10-2y}{8}$$

$$\begin{aligned} \text{Then } E(X) &= \int_0^2 x f_X(x) dx = \int_0^2 x \left(\frac{6-2x}{8} \right) dx \\ &= \frac{1}{8} \left[\frac{6x^2}{2} - \frac{2x^3}{3} \right]_0^2 \\ &= \frac{1}{8} \left[12 - \frac{16}{3} \right] = \frac{1}{8} \times \frac{20}{3} = \frac{5}{6} \end{aligned}$$

$$E(Y) = \int_2^4 y \left(\frac{10-2y}{8} \right) dy = \frac{1}{8} \left[\frac{10y^2}{2} - \frac{2y^3}{3} \right]_2^4 = \frac{17}{6}$$

$$E(X^2) = \int_0^2 x^2 f_X(x) dx = \int_0^2 x^2 \left(\frac{6-2x}{8} \right) dx = \frac{1}{8} \left[\frac{6x^3}{3} - \frac{2x^4}{4} \right]_0^2 = 1$$

$$E(Y^2) = \int_2^4 y^2 \left(\frac{10-2y}{8} \right) dy = \frac{1}{8} \left[\frac{10y^3}{3} - \frac{2y^4}{4} \right]_2^4 = \frac{25}{3}$$

$$Var(X) = \sigma_X^2 = E(X^2) - [E(X)]^2 = 1 - \left(\frac{5}{6} \right)^2 = \frac{11}{36}$$

$$Var(Y) = \sigma_Y^2 = E(Y^2) - [E(Y)]^2 = \frac{25}{3} - \left(\frac{17}{6} \right)^2 = \frac{11}{36}$$

$$\begin{aligned} E(XY) &= \int_2^4 \int_0^2 xy \left(\frac{6-x-y}{8} \right) dx dy \\ &= \frac{1}{8} \int_2^4 \left[\frac{6x^2 y}{2} - \frac{x^3 y}{3} - \frac{x^2 y^2}{2} \right]_0^2 dy \\ &= \frac{1}{8} \int_2^4 \left(12y - \frac{8}{3}y - 2y^2 \right) dy = \frac{1}{8} \left[\frac{12y^2}{2} - \frac{8}{3} \frac{y^2}{2} - \frac{2y^3}{3} \right]_2^4 \\ &= \frac{1}{8} \left[96 - \frac{64}{3} - \frac{128}{3} - 24 + \frac{16}{3} + \frac{16}{3} \right] = \frac{1}{8} \left[\frac{56}{3} \right] \end{aligned}$$

$$E(XY) = \frac{7}{3}$$

$$\rho_{XY} = \frac{E(XY) - E(X)E(Y)}{\sigma_X \sigma_Y} = \frac{\frac{7}{3} - \left(\frac{5}{6}\right)\left(\frac{17}{6}\right)}{\frac{\sqrt{11}}{6} \frac{\sqrt{11}}{6}}$$

$$\rho_{XY} = -\frac{1}{11}.$$

22.a). Let the random variables X and Y have pdf $f(x,y)=\frac{1}{3}, (x,y)=(0,0),(1,1),(2,0)$.

Compute the correlation coefficient.

b) Let X_1 and X_2 be two independent random variables with means 5 and 10 and standard devotions 2 and 3 respectively. Obtain the correlation coefficient of UV where $U=3X_1+4X_2$ and $V=3X_1-X_2$.

Solution:

a). The probability distribution is

$\begin{matrix} X \\ Y \end{matrix}$	0	1	2	$P(Y)$
0	$\frac{1}{3}$	0	0	$\frac{1}{3}$
1	0	$\frac{1}{3}$	0	$\frac{1}{3}$
0	0	0	$\frac{1}{3}$	$\frac{1}{3}$
$P(X)$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	

$$E(X)=\sum_i x_i p_i(x_i)=\left(0\times\frac{1}{3}\right)+\left(1\times\frac{1}{3}\right)+\left(2\times\frac{1}{3}\right)=1$$

$$E(Y)=\sum_j y_j p_j(y_j)=\left(0\times\frac{1}{3}\right)+\left(1\times\frac{1}{3}\right)+\left(0\times\frac{1}{3}\right)=\frac{1}{3}$$

$$E(X^2)=\sum_i x_i^2 p(x_i)=\left(0\times\frac{1}{3}\right)+\left(1\times\frac{1}{3}\right)+\left(4\times\frac{1}{3}\right)=\frac{5}{3}$$

$$Var(X)=E(X^2)-[E(X)]^2=\frac{5}{3}-1=\frac{2}{3}$$

$$E(Y^2)=\sum_j y_j^2 p(y_j)=\left(0\times\frac{1}{3}\right)+\left(1\times\frac{1}{3}\right)+\left(0\times\frac{1}{3}\right)=\frac{1}{3}$$

$$\therefore V(Y)=E(Y^2)-[E(Y)]^2=\frac{1}{3}-\frac{1}{9}=\frac{2}{9}$$

$$\text{Correlation coefficient } \rho_{XY} = \frac{E(XY) - E(X)E(Y)}{\sqrt{V(X)V(Y)}}$$

$$E(XY) = \sum_i \sum_j x_i y_j p(x_i, y_j)$$

$$= 0.0 \cdot \frac{1}{3} + 0.1 \cdot 0 + 1.0 \cdot 0 + 1.1 \cdot \frac{1}{3} + 1.2 \cdot 0 + 0.0 \cdot 0 + 0.1 \cdot 0 + 0.2 \cdot \frac{1}{3} = \frac{1}{3}$$

$$\rho_{XY} = \frac{\frac{1}{3} - (1)\left(\frac{1}{3}\right)}{\sqrt{\frac{2}{3} \times \frac{2}{9}}} = 0$$

Correlation coefficient = 0.

b). Given $E(X_1) = 5$, $E(X_2) = 10$

$$V(X_1) = 4, \quad V(X_2) = 9$$

Since X and Y are independent $E(XY) = E(X)E(Y)$

$$\text{Correlation coefficient} = \frac{E(UV) - E(U)E(V)}{\sqrt{\text{Var}(U)\text{Var}(V)}}$$

$$\begin{aligned} E(U) &= E(3X_1 + 4X_2) = 3E(X_1) + 4E(X_2) \\ &= (3 \times 5) + (4 \times 10) = 15 + 40 = 55. \end{aligned}$$

$$\begin{aligned} E(V) &= E(3X_1 - X_2) = 3E(X_1) - E(X_2) \\ &= (3 \times 5) - 10 = 15 - 10 = 5 \end{aligned}$$

$$\begin{aligned} E(UV) &= E[(3X_1 + 4X_2)(3X_1 - X_2)] \\ &= E[9X_1^2 - 3X_1X_2 + 12X_1X_2 - 4X_2^2] \\ &= 9E(X_1^2) - 3E(X_1X_2) + 12E(X_1X_2) - 4E(X_2^2) \\ &= 9E(X_1^2) + 9E(X_1X_2) - 4E(X_2^2) \\ &= 9E(X_1^2) + 9E(X_1)E(X_2) - 4E(X_2^2) \\ &= 9E(X_1^2) + 450 - 4E(X_2^2) \end{aligned}$$

$$V(X_1) = E(X_1^2) - [E(X_1)]^2$$

$$E(X_1^2) = V(X_1) + [E(X_1)]^2 = 4 + 25 = 29$$

$$E(X_2^2) = V(X_2) + [E(X_2)]^2 = 9 + 100 = 109$$

$$\begin{aligned} \therefore E(UV) &= (9 \times 29) + 450 - (4 \times 109) \\ &= 261 + 450 - 436 = 275 \end{aligned}$$

$$\text{Cov}(U, V) = E(UV) - E(U)E(V)$$

$$= 275 - (5 \times 55) = 0$$

Since $Cov(U, V) = 0$, Correlation coefficient = 0.

23.a). Let the random variable X has the marginal density function $f(x) = 1, -\frac{1}{2} < x < \frac{1}{2}$ and

let the conditional density of Y be $f\left(\frac{y}{x}\right) = \begin{cases} 1, & x < y < x+1, & -\frac{1}{2} < x < 0 \\ 1, & -x < y < 1-x, & 0 < x < \frac{1}{2} \end{cases}$. Prove that the

variables X and Y are uncorrelated.

b). Given $f(x, y) = xe^{-x(y+1)}, x \geq 0, y \geq 0$. Find the regression curve of Y on X .

Solution:

$$\text{a). We have } E(X) = \int_{-\frac{1}{2}}^{\frac{1}{2}} xf(x)dx = \int_{-\frac{1}{2}}^{\frac{1}{2}} xdx = \left[\frac{x^2}{2} \right]_{-\frac{1}{2}}^{\frac{1}{2}} = 0$$

$$\begin{aligned} E(XY) &= \int_{-\frac{1}{2}}^0 \int_x^{x+1} xy dx dy + \int_0^{\frac{1}{2}} \int_{-x}^{1-x} xy dx dy \\ &= \int_{-\frac{1}{2}}^0 x \left[\int_x^{x+1} y dy \right] dx + \int_0^{\frac{1}{2}} x \left[\int_{-x}^{1-x} y dy \right] dx \\ &= \frac{1}{2} \int_{-\frac{1}{2}}^0 x(2x+1) dx + \frac{1}{2} \int_0^{\frac{1}{2}} x(1-2x) dx \\ &= \frac{1}{2} \left[\frac{2x^3}{3} + \frac{x^2}{2} \right]_{-\frac{1}{2}}^0 + \frac{1}{2} \left[\frac{x^2}{2} - \frac{2x^3}{3} \right]_0^{\frac{1}{2}} = 0 \end{aligned}$$

Since $Cov(X, Y) = E(XY) - E(X)E(Y) = 0$, the variables X and Y are uncorrelated.

b). Regression curve of Y on X is $E\left(\frac{y}{x}\right)$

$$\begin{aligned} E\left(\frac{y}{x}\right) &= \int_{-\infty}^{\infty} y f\left(\frac{y}{x}\right) dy \\ f(y/x) &= \frac{f(x, y)}{f_X(X)} \end{aligned}$$

Marginal density function $f_X(x) = \int_0^{\infty} f(x, y) dy$

$$= x \int_0^{\infty} e^{-x(y+1)} dy$$

$$= x \left[\frac{e^{-x(y+1)}}{-x} \right]_0^{\infty} = e^{-x}, \quad x \geq 0$$

Conditional pdf of Y on X is $f\left(\frac{y}{x}\right) = \frac{f(x, y)}{f_X(x)} = \frac{xe^{-xy-x}}{e^{-x}} = xe^{-xy}$

The regression curve of Y on X is given by

$$E\left(\frac{y}{x}\right) = \int_0^{\infty} yxe^{-xy} dy$$

$$= x \left[y \frac{e^{-xy}}{-x} - \frac{e^{-xy}}{x^2} \right]_0^{\infty}$$

$$E\left(\frac{y}{x}\right) = \frac{1}{x} \Rightarrow y = \frac{1}{x} \text{ and hence } xy = 1.$$

24.a). Given $f(x, y) = \begin{cases} \frac{x+y}{3}, & 0 < x < 1, 0 < y < 2 \\ 0 & , \text{otherwise} \end{cases}$, obtain the regression of Y on X and X on

Y .

b). Distinguish between correlation and regression Analysis

Solution:

a). Regression of Y on X is $E\left(\frac{Y}{X}\right)$

$$E\left(\frac{Y}{X}\right) = \int_{-\infty}^{\infty} yf\left(\frac{y}{x}\right) dy$$

$$f\left(\frac{Y}{X}\right) = \frac{f(x, y)}{f_X(x)}$$

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

$$= \int_0^2 \left(\frac{x+y}{3}\right) dy = \frac{1}{3} \left[xy + \frac{y^2}{2} \right]_0^2$$

$$= \frac{2(x+1)}{3}$$

$$f\left(\frac{Y}{X}\right) = \frac{f(x, y)}{f_X(x)} = \frac{x+y}{2(x+1)}$$

$$\text{Regression of } Y \text{ on } X = E\left(\frac{Y}{X}\right) = \int_0^2 \frac{y(x+y)}{2(x+1)} dy$$

$$= \frac{1}{2(x+1)} \left[\frac{xy^2}{2} + \frac{y^3}{3} \right]_0^1$$

$$= \frac{1}{2(x+1)} \left[2x + \frac{8}{3} \right] = \frac{3x+4}{3(x+1)}$$

$$E\left(\frac{X}{Y}\right) = \int_{-\infty}^{\infty} xf\left(\frac{x}{y}\right)dx$$

$$f\left(\frac{x}{y}\right) = \frac{f(x,y)}{f_Y(y)}$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)dx$$

$$= \int_0^1 \left(\frac{x+y}{3} \right) dx = \frac{1}{3} \left[\frac{x^2}{2} + xy \right]_0^1$$

$$= \frac{1}{3} \left[\frac{1}{2} + y \right]$$

$$f\left(\frac{x}{y}\right) = \frac{2(x+y)}{2y+1}$$

$$\text{Regression of } X \text{ on } Y = E\left(\frac{X}{Y}\right) = \int_0^1 \frac{x+y}{2y+1} dx$$

$$= \frac{1}{2y+1} \left[\frac{x^2}{2} + xy \right]_0^1$$

$$= \frac{\frac{1}{2} + y}{2y+1} = \frac{1}{2}.$$

b).

1. Correlation means relationship between two variables and Regression is a Mathematical Measure of expressing the average relationship between the two variables.
2. Correlation need not imply cause and effect relationship between the variables. Regression analysis clearly indicates the cause and effect relationship between Variables.
3. Correlation coefficient is symmetric i.e. $r_{xy} = r_{yx}$ where regression coefficient is not symmetric
4. Correlation coefficient is the measure of the direction and degree of linear relationship between two variables. In regression using the relationship between two variables we can predict the dependent variable value for any given independent variable value.

25.a). X any Y are two random variables with variances σ_x^2 and σ_y^2 respectively and r is the coefficient of correlation between them. If $U = X + KY$ and $V = X + \frac{y\sigma_x}{\sigma_y}$, find the value of k so that U and V are uncorrelated.

b). Find the regression lines:

X	6	8	10	18	20	23
Y	40	36	20	14	10	2

Solution:

Given $U = X + KY$

$$E(U) = E(X) + KE(Y)$$

$$V = X + \frac{\sigma_X}{\sigma_Y} Y$$

$$E(V) = E(X) + \frac{\sigma_X}{\sigma_Y} E(Y)$$

If U and V are uncorrelated, $Cov(U, V) = 0$

$$E[(U - E(U))(V - E(V))] = 0$$

$$\Rightarrow E\left[(X + KY - E(X) - KE(Y)) \times \left(X + \frac{\sigma_X}{\sigma_Y} Y - E(X) - \frac{\sigma_X}{\sigma_Y} E(Y)\right)\right] = 0$$

$$\Rightarrow E\left\{\left[(X - E(X)) + K(Y - E(Y))\right] \times \left[(X - E(X)) + \frac{\sigma_X}{\sigma_Y}(Y - E(Y))\right]\right\} = 0$$

$$\Rightarrow E\left\{(X - E(X))^2 + \frac{\sigma_X}{\sigma_Y}(X - E(X))(Y - E(Y)) + K(Y - E(Y))(X - E(X)) + K \frac{\sigma_X}{\sigma_Y}(Y - E(Y))^2\right\} = 0$$

$$\Rightarrow V(X) + \frac{\sigma_X}{\sigma_Y} Cov(X, Y) + KCov(X, Y) + K \frac{\sigma_X}{\sigma_Y} V(Y) = 0$$

$$K\left[Cov(X, Y) + \frac{\sigma_X}{\sigma_Y} V(Y)\right] = -V(X) - \frac{\sigma_X}{\sigma_Y} Cov(x, y)$$

$$K = \frac{-V(X) - \frac{\sigma_X}{\sigma_Y} r\sigma_X\sigma_Y}{r\sigma_X\sigma_Y + \frac{\sigma_X}{\sigma_Y} V(Y)} = \frac{-\sigma_X^2 - r\sigma_X^2}{r\sigma_X\sigma_Y + \sigma_X\sigma_Y}$$

$$= \frac{-\sigma_X^2(1+r)}{\sigma_X\sigma_Y(1+r)} = -\frac{\sigma_X}{\sigma_Y}$$

b).

X	Y	X^2	Y^2	XY
6	40	36	1600	240

8	36	64	1296	288
10	20	100	400	200
18	14	324	196	252
20	10	400	100	200
23	2	529	4	46
$\sum X = 85$	$\sum Y = 122$	$\sum X^2 = 1453$	$\sum Y^2 = 3596$	$\sum XY = 1226$

$$\bar{X} = \frac{\sum x}{n} = \frac{85}{6} = 14.17, \bar{Y} = \frac{\sum y}{n} = \frac{122}{6} = 20.33$$

$$\sigma_x = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2} = \sqrt{\frac{1453}{6} - \left(\frac{85}{6}\right)^2} = 6.44$$

$$\sigma_y = \sqrt{\frac{\sum y^2}{n} - \left(\frac{\sum y}{n}\right)^2} = \sqrt{\frac{3596}{6} - \left(\frac{122}{6}\right)^2} = 13.63$$

$$r = \frac{\frac{\sum xy}{n} - \bar{x}\bar{y}}{\sigma_x \sigma_y} = \frac{\frac{1226}{6} - (14.17)(20.33)}{(6.44)(13.63)} = -0.95$$

$$b_{xy} = r \frac{\sigma_x}{\sigma_y} = -0.95 \times \frac{6.44}{13.63} = -0.45$$

$$b_{yx} = r \frac{\sigma_y}{\sigma_x} = -0.95 \times \frac{13.63}{6.44} = -2.01$$

The regression line X on Y is

$$x - \bar{x} = b_{xy} (y - \bar{y}) \Rightarrow x - 14.17 = -0.45 (y - 20.33)$$

$$\Rightarrow x = -0.45y + 23.32$$

The regression line Y on X is

$$y - \bar{y} = b_{yx} (x - \bar{x}) \Rightarrow y - 20.33 = -2.01 (x - 14.17)$$

$$\Rightarrow y = -2.01x + 48.81$$

26. a) Using the given information given below compute $\bar{x}, \bar{y}$ and r . Also compute σ_y when $\sigma_x = 2, 2x + 3y = 8$ and $4x + y = 10$.

b) The joint pdf of X and Y is

	X	
Y	-1	1

0	$\frac{1}{8}$	$\frac{3}{8}$
1	$\frac{2}{8}$	$\frac{2}{8}$

Find the correlation coefficient of X and Y .

Solution:

a). When the regression equation are Known the arithmetic means are computed by solving the equation.

$$2x + 3y = 8 \text{ ----- (1)}$$

$$4x + y = 10 \text{ ----- (2)}$$

$$(1) \times 2 \Rightarrow 4x + 6y = 16 \text{ ----- (3)}$$

$$(2) - (3) \Rightarrow -5y = -6$$

$$\Rightarrow y = \frac{6}{5}$$

$$\text{Equation (1)} \Rightarrow 2x + 3\left(\frac{6}{5}\right) = 8$$

$$\Rightarrow 2x = 8 - \frac{18}{5}$$

$$\Rightarrow x = \frac{11}{5}$$

$$\text{i.e. } \bar{x} = \frac{11}{5} \text{ \& } \bar{y} = \frac{6}{5}$$

To find r , Let $2x + 3y = 8$ be the regression equation of X on Y .

$$2x = 8 - 3y \Rightarrow x = 4 - \frac{3}{2}y$$

$$\Rightarrow b_{xy} = \text{Coefficient of } Y \text{ in the equation of } X \text{ on } Y = -\frac{3}{2}$$

Let $4x + y = 10$ be the regression equation of Y on X

$$\Rightarrow y = 10 - 4x$$

$$\Rightarrow b_{yx} = \text{coefficient of } X \text{ in the equation of } Y \text{ on } X = -4.$$

$$\begin{aligned} r &= \pm \sqrt{b_{xy} b_{yx}} \\ &= -\sqrt{\left(-\frac{3}{2}\right)(-4)} \quad \left(\because b_{xy} \text{ \& } b_{yx} \text{ are negative}\right) \\ &= -2.45 \end{aligned}$$

Since r is not in the range of $(-1 \leq r \leq 1)$ the assumption is wrong.

Now let equation (1) be the equation of Y on X

$$\Rightarrow y = \frac{8}{3} - \frac{2x}{3}$$

$$\Rightarrow b_{yx} = \text{Coefficient of } X \text{ in the equation of } Y \text{ on } X$$

$$b_{yx} = -\frac{2}{3}$$

from equation (2) be the equation of X on Y

$$b_{xy} = -\frac{1}{4}$$

$$r = \pm \sqrt{b_{xy} b_{yx}} = \sqrt{-\frac{2}{3} \times -\frac{1}{4}} = 0.4081$$

To compute σ_y from equation (4) $b_{yx} = -\frac{2}{3}$

$$\text{But we know that } b_{yx} = r \frac{\sigma_y}{\sigma_x}$$

$$\Rightarrow -\frac{2}{3} = 0.4081 \times \frac{\sigma_y}{2}$$

$$\Rightarrow \sigma_y = -3.26$$

b). Marginal probability mass function of X is

$$\text{When } X = 0, P(X) = \frac{1}{8} + \frac{3}{8} = \frac{4}{8}$$

$$X = 1, P(X) = \frac{2}{8} + \frac{2}{8} = \frac{4}{8}$$

Marginal probability mass function of Y is

$$\text{When } Y = -1, P(Y) = \frac{1}{8} + \frac{2}{8} = \frac{3}{8}$$

$$Y = 1, P(Y) = \frac{3}{8} + \frac{2}{8} = \frac{5}{8}$$

$$E(X) = \sum_x x p(x) = 0 \times \frac{4}{8} + 1 \times \frac{4}{8} = \frac{4}{8}$$

$$E(Y) = \sum_y y p(y) = -1 \times \frac{3}{8} + 1 \times \frac{5}{8} = -\frac{3}{8} + \frac{5}{8} = \frac{2}{8}$$

$$E(X^2) = \sum_x x^2 p(x) = 0^2 \times \frac{4}{8} + 1^2 \times \frac{4}{8} = \frac{4}{8}$$

$$E(Y^2) = \sum_y y^2 p(y) = (-1)^2 \times \frac{3}{8} + 1^2 \times \frac{5}{8} = \frac{3}{8} + \frac{5}{8} = 1$$

$$V(X) = E(X^2) - (E(X))^2$$

$$= \frac{4}{8} - \left(\frac{4}{8}\right)^2 = \frac{1}{4}$$

$$V(Y) = E(Y^2) - (E(Y))^2$$

$$= 1 - \left(\frac{1}{4}\right)^2 = \frac{15}{16}$$

$$\begin{aligned}
E(XY) &= \sum_x \sum_y xy p(x,y) \\
&= 0 \times \frac{1}{8} + 0 \times \frac{3}{8} + (-1) \frac{2}{8} + 1 \times \left(\frac{2}{8}\right) = 0 \\
Cov(X,Y) &= E(XY) - E(X)E(Y) = 0 - \frac{1}{2} \times \frac{1}{4} = -\frac{1}{8} \\
r &= \frac{Cov(X,Y)}{\sqrt{V(X)}\sqrt{V(Y)}} = \frac{-\frac{1}{8}}{\sqrt{\frac{1}{4}}\sqrt{\frac{15}{16}}} = -0.26.
\end{aligned}$$

27. a) Calculate the correlation coefficient for the following heights (in inches) of fathers X and their sons Y .

X	65	66	67	67	68	69	70	72
Y	67	68	65	68	72	72	69	71

b) If X and Y are independent exponential variates with parameters 1, find the pdf of $U = X - Y$.

Solution:

X	Y	XY	X^2	Y^2
65	67	4355	4225	4489
66	68	4488	4359	4624
67	65	4355	4489	4285
68	72	4896	4624	5184
69	72	4968	4761	5184
70	69	4830	4900	4761
72	71	5112	5184	5041
$\sum X = 544$	$\sum Y = 552$	$\sum XY = 37560$	$\sum X^2 = 37028$	$\sum Y^2 = 38132$

$$\begin{aligned}
\overline{X} &= \frac{\sum x}{n} = \frac{544}{8} = 68 \\
\overline{Y} &= \frac{\sum y}{n} = \frac{552}{8} = 69 \\
\overline{XY} &= 68 \times 69 = 4692 \\
\sigma_x &= \sqrt{\frac{1}{n} \sum x^2 - \overline{X}^2} = \sqrt{\frac{1}{8} (37028) - 68^2} = \sqrt{4628.5 - 4624} = 2.121 \\
\sigma_y &= \sqrt{\frac{1}{n} \sum y^2 - \overline{y}^2} = \sqrt{\frac{1}{8} (38132) - 69^2} = \sqrt{4766.5 - 4761} = 2.345
\end{aligned}$$

$$\begin{aligned} \text{Cov}(X, Y) &= \frac{1}{n} \sum XY - \bar{X} \bar{Y} = \frac{1}{8} (37650) - 68 \times 69 \\ &= 4695 - 4692 = 3 \end{aligned}$$

The correlation coefficient of X and Y is given by

$$\begin{aligned} r(X, Y) &= \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{3}{(2.121)(2.345)} \\ &= \frac{3}{4.973} = 0.6032. \end{aligned}$$

b). Given that X and Y are exponential variates with parameters 1

$$f_X(x) = e^{-x}, x \geq 0, f_Y(y) = e^{-y}, y \geq 0$$

Also $f_{XY}(x, y) = f_X(x) f_Y(y)$ since X and Y are independent

$$\begin{aligned} &= e^{-x} e^{-y} \\ &= e^{-(x+y)}; x \geq 0, y \geq 0 \end{aligned}$$

Consider the transformations $u = x - y$ and $v = y$

$$\Rightarrow x = u + v, y = v$$

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = 1$$

$$\begin{aligned} f_{UV}(u, v) &= f_{XY}(x, y) \left| \frac{\partial u}{\partial x} \frac{\partial v}{\partial y} \right| J = e^{-x} e^{-y} = e^{-(u+v)} e^{-v} \\ &= e^{-(u+2v)}, u + v \geq 0, v \geq 0 \end{aligned}$$

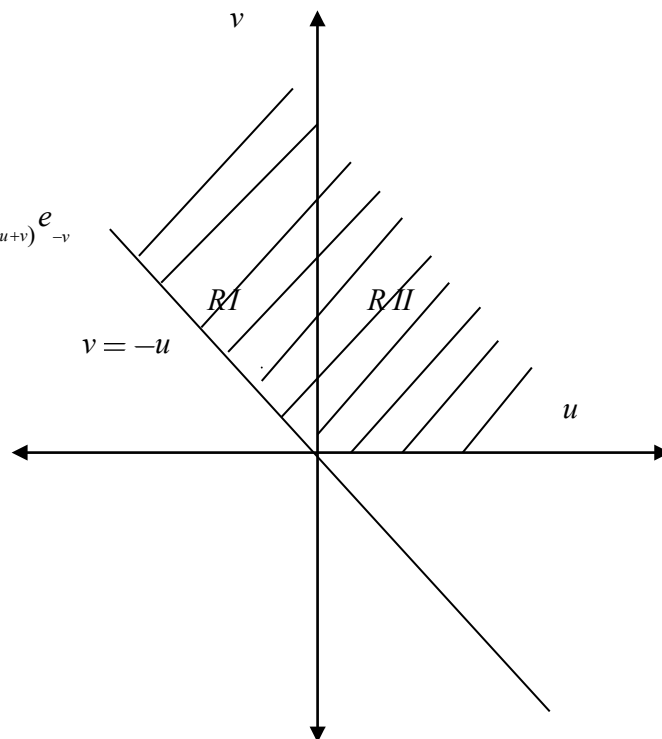
In Region I when $u < 0$

$$\begin{aligned} f(u) &= \int_{-u}^{\infty} f(u, v) dv = \int_{-u}^{\infty} e^{-u} \cdot e^{-2v} dv \\ &= e^{-u} \left[\frac{e^{-2v}}{-2} \right]_{-u}^{\infty} \\ &= \frac{e^{-u}}{-2} [0 - e^{2u}] = \frac{e^u}{2} \end{aligned}$$

In Region II when $u > 0$

$$\begin{aligned} f(u) &= \int_0^{\infty} f(u, v) dv \\ &= \int_0^{\infty} e^{-(u+2v)} dv = \frac{e^{-u}}{2} \end{aligned}$$

$$\therefore f(u) = \begin{cases} \frac{e^u}{2}, & u < 0 \\ \frac{e^{-u}}{2}, & u > 0 \end{cases}$$



28. a) The joint pdf of X and Y is given by $f(x, y) = e^{-(x+y)}, x > 0, y > 0$. Find the pdf of $U = \frac{X+Y}{2}$.

b) If X and Y are independent random variables each following $N(0, 2)$, find the pdf of $Z = 2X + 3Y$. If X and Y are independent rectangular variates on $(0, 1)$ find the distribution of $\frac{X}{Y}$.

Solution:

a). Consider the transformation $u = \frac{x+y}{2}$ & $v = y$
 $\Rightarrow x = 2u - v$ and $y = v$

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 2 & -1 \\ 0 & 1 \end{vmatrix} = 2$$

$$\begin{aligned} f_{UV}(u, v) &= f_{XY}(x, y) |J| \\ &= e^{-(x+y)} 2 = 2e^{-(x+y)} = 2e^{-(2u-v+v)} \\ &= 2e^{-2u}, \quad 2u - v \geq 0, \quad v \geq 0 \end{aligned}$$

$$f_{UV}(u, v) = 2e^{-2u}, \quad u \geq 0, \quad 0 \leq v \leq \frac{u}{2}$$

$$f(u) = \int_0^{\frac{u}{2}} f_{UV}(u, v) dv = \int_0^{\frac{u}{2}} 2e^{-2u} dv$$

$$= \left[2e^{-2u} v \right]_0^{\frac{u}{2}}$$

$$f(u) = \begin{cases} 2 \frac{u}{2} e^{-2u}, & u \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

b).(i) Consider the transformations $w = y$,

i.e. $z = 2x + 3y$ and $w = y$

i.e. $x = \frac{1}{2}(z - 3w), y = w$

$$|J| = \frac{\partial(x, y)}{\partial(z, w)} = \begin{vmatrix} \frac{\partial x}{\partial z} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial z} & \frac{\partial y}{\partial w} \end{vmatrix} = \begin{vmatrix} \frac{1}{2} & -\frac{3}{2} \\ 0 & 1 \end{vmatrix} = \frac{1}{2}.$$

Given that X and Y are independent random variables following $N(0, 2)$

$$\therefore f_{XY}(x, y) = \frac{1}{8\pi} e^{-\frac{(x^2+y^2)}{8}}, -\infty < x, y < \infty$$

The joint pdf of (z, w) is given by

$$\begin{aligned} f_{ZW}(z, w) &= |J| f_{XY}(x, y) \\ &= \frac{1}{2} \cdot \frac{1}{8\pi} e^{-\frac{\left[\frac{1}{4}(z-3w)^2 + w^2\right]}{8}} \\ &= \frac{1}{16\pi} e^{-\frac{1}{32}[(z-3w)^2 + 4w^2]}, -\infty < z, w < \infty. \end{aligned}$$

The pdf of z is the marginal pdf obtained by interchanging $f_{ZW}(z, w)$ w.r.to w over the range of w .

$$\begin{aligned} \therefore f_Z(z) &= \frac{1}{16\pi} \int_{-\infty}^{\infty} \left(e^{-\frac{1}{32}(z^2 - 6wz + 13w^2)} \right) dw \\ &= \frac{1}{16\pi} e^{-\frac{z^2}{32}} \int_{-\infty}^{\infty} \left(e^{-\frac{13}{32} \left(w^2 - \frac{6wz}{13} + \left(\frac{3z}{13} \right)^2 - \left(\frac{3z}{13} \right)^2 \right)} \right) dw \\ &= \frac{1}{16\pi} e^{-\frac{z^2}{32} + \frac{9z^2}{13 \times 32}} \int_{-\infty}^{\infty} \left(e^{-\frac{13}{32} \left(w - \frac{3z}{13} \right)^2} \right) dw \\ &= \frac{1}{16\pi} e^{-\frac{z^2}{8 \times 13}} \int_{-\infty}^{\infty} e^{-\frac{13}{32} t^2} dt \\ r = \frac{13}{32} t^2 \Rightarrow dr &= \frac{13}{16} t dt \Rightarrow \frac{16}{13t} dr = dt \Rightarrow \sqrt{\frac{r32}{13}} dr = dt \\ \frac{16}{13} \sqrt{\frac{13}{r32}} dr &= dt \Rightarrow \frac{4}{\sqrt{13} \times \sqrt{2}} r^{-\frac{1}{2}} dr = dt \\ &= \frac{2}{16\pi} \frac{4}{\sqrt{13} \times \sqrt{2}} e^{-\frac{z^2}{8 \times 13}} \int_0^{\infty} e^{-r} r^{-\frac{1}{2}} dr \\ &= \frac{1}{2\pi \sqrt{13} \times \sqrt{2}} e^{-\frac{z^2}{8 \times 13}} \int_0^{\infty} e^{-r} r^{-\frac{1}{2}} dr \\ &= \frac{1}{2\pi \sqrt{13} \times \sqrt{2}} e^{-\frac{z^2}{8 \times 13}} \sqrt{\pi} = \frac{1}{2\sqrt{13} \sqrt{2\pi}} e^{-\frac{z^2}{2(2\sqrt{13})^2}} \end{aligned}$$

$$\text{i.e. } Z \sim N(0, 2\sqrt{13})$$

b).(ii) Given that X and Y are uniform Variants over $(0,1)$

$$\therefore f_X(x) = \begin{cases} 1, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases} \text{ and } f_Y(y) = \begin{cases} 1, & 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

Since X and Y are independent,

$$f_{XY}(x, y) = f_X(x) f_Y(y) \begin{cases} 1, & 0 < x, y < 1 \\ 0, & \text{otherwise} \end{cases}$$

Consider the transformation $u = \frac{x}{y}$ and $v = y$

i.e. $x = uv$ and $y = v$

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} v & 0 \\ u & 1 \end{vmatrix} = v$$

$$\begin{aligned} \therefore f_{UV}(u, v) &= f_{XY}(x, y) |J| \\ &= v, \quad 0 < u < \infty, \quad 0 < v < \infty \end{aligned}$$

The range for u and v are identified as follows.

$$0 < x < 1 \text{ and } 0 < y < 1.$$

$$\Rightarrow 0 < uv < 1 \text{ and } 0 < v < 1$$

$$\Rightarrow uv > 0, uv < 1, v > 0 \text{ and } v < 1$$

$$\Rightarrow uv > 0 \text{ and } v > 0 \Rightarrow u > 0$$

$$\text{Now } f(u) = \int f_{UV}(u, v) dv$$

The range for v differs in two regions

$$\begin{aligned} f(u) &= \int_0^1 f_{UV}(u, v) dv \\ &= \int_0^1 v dv = \left[\frac{v^2}{2} \right]_0^1 = \frac{1}{2}, \quad 0 < u < 1 \end{aligned}$$

$$\begin{aligned} f(u) &= \int_0^{\frac{1}{u}} f_{UV}(u, v) dv \\ &= \int_0^{\frac{1}{u}} v dv = \left[\frac{v^2}{2} \right]_0^{\frac{1}{u}} = \frac{1}{2u^2}, \quad 1 \leq u < \infty \end{aligned}$$

$$\therefore f(u) = \begin{cases} \frac{1}{2}, & 0 \leq u \leq 1 \\ \frac{1}{2u^2}, & u > 1 \end{cases}$$

Probability and Statistics
Analysis of Variance and Statistical Quality Control

ANOVA (Analysis of Variance) :

Analysis of Variance is a technique that will enable us to test for the significance of the difference among more than two sample means.

Assumptions of analysis of variance:

- (i) The sample observations are independent
- (ii) The environmental effects are additive in nature
- (iii) The samples have been randomly selected from the population.
- (iv) Parent population from which observations are taken is normal.

One Way Classification (or) Completely randomized Design (C.R.D)

The C.R.D is the simplest of all the designs, based on principles of randomization and replication. In this design, treatments are allocated at random to the experimental units over the entire experimental materials.

Advantages of completely randomized block design:

The advantages of completely randomized experimental design are as follows:

- (i) Easy to lay out.
- (ii) Allow flexibility
- (iii) Simple statistical analysis
- (iv) Lots of information due to missing data is smaller than with any other design

Two Way Classification (or) Randomized Block Design (R.B.D):

The entire experiment influences on only two factors is two way Classification.

The basic principles of design of experiments:

- (i) Randomization
- (ii) Replication
- (iii) Local Control

Working Procedure (One – Way classification)

Null Hypothesis H_0 : There is no significance difference between the treatments.

Alternate Hypothesis H_1 : There is a significance difference between the treatments.

Analysis:

Step 1: Find N = number of observations

Step 2: Find T = The total value of observations

Step 3: Find the correction Factor = $C.F = \frac{T^2}{N}$

Step 4: Calculate the total sum of squares = $TSS = (\sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \dots) - C.F$

Step 4: Find Total Sum of Square $TSS = (\sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \dots) - C.F$

Step 5: Column Sum of Square $SSC=\left\{\frac{(\sum X_1)^2}{N_1}+\frac{(\sum X_2)^2}{N_2}+\frac{(\sum X_3)^2}{N_3}+...\right\}-C.F$

Where N_i =Total number of observation in each column ($i = 1,2,3,...$)

Step 6: Prepare the ANOVA TABLE to calculate F-ratio.

Source of Variation	Sum of Degrees	Degree of freedom	Mean Square	F- Ratio
Between Columns	SSC	c-1	$MSC = \frac{SSC}{c-1}$	$F_c = \frac{MSC}{MSE}$ if $MSC > MSE$ (or) $F_c = \frac{MSE}{MSC}$ if $MSE > MSC$
Error	SSE	N-c	$MSE = \frac{SSE}{N-c}$	
Total				

Step 7: Find the table value (use χ^2 table)

Step 8: Conclusion:

Calculated value < Table Value, the we accept Null Hypothesis H_0 (or)

Calculated value > Table Value, the we rejectNull Hypothesis H_0

Working Procedure (Two – Way classification)

Null Hypothesis H_0 : There is no significance difference between the treatments.

Alternate Hypothesis H_1 : There is a significance difference between the treatments.

Analysis:

Step 1: Find N = number of observations

Setp 2: Find T = The total value of observations

Step 3: Find the correction Factor = $C.F = \frac{T^2}{N}$

Step 4: Calculate the total sum of squares = $TSS= (\sum X_1^2 + \sum X_2^2 + \sum X_3^2 +...)-C.F$

Step 4: Find Total Sum of Square $TSS= (\sum X_1^2 + \sum X_2^2 + \sum X_3^2 +...)-C.F$

Step 5: Find column sum of Square $SSC=\left\{\frac{(\sum X_1)^2}{N_1}+\frac{(\sum X_2)^2}{N_2}+\frac{(\sum X_3)^2}{N_3}+...\right\}-C.F$

Where N_i =Total number of observation in each column ($i = 1,2,3,...$)

Step 6: Find Row sum of square = $SSR = \left\{ \frac{(\sum Y_1)^2}{N_1} + \frac{(\sum Y_2)^2}{N_2} + \frac{(\sum Y_3)^2}{N_3} + \dots \right\} - C.F$

Where N_j =Total number of observation in each Row ($j = 1,2,3,\dots$)

Step 7:Prepare the ANOVA TABLE to calculate F-ratio.

Source of Variation	Sum of Degrees	Degree of freedom	Mean Square	F- Ratio
Between Columns	SSC	c-1	$MSC = \frac{SSC}{c - 1}$	$F_c = \frac{MSC}{MSE}$ if $MSC > MSE$ (or) $F_c = \frac{MSE}{MSC}$ if $MSE > MSC$
Between Rows	SSR	r-1	$MSC = \frac{SSR}{r - 1}$	$F_r = \frac{MSR}{MSE}$ if $MSR > MSE$ (or) $F_r = \frac{MSE}{MSR}$ if $MSE > MSR$
Error	SSE	N-c-r+1	$MSE = \frac{SSE}{N - c - r + 1}$	
Total	TSS	rc-1		

Step 8: Find the table value for both F_C & F_R (use χ^2 table)

Step 9:Conclusion:

Calculated value < Table Value, the we accept Null Hypothesis H_0 (or)

Calculated value > Table Value, the we reject Null Hypothesis H_0

1

The following are the numbers of mistakes made in 5 successive days of 4 technicians working for a photographic laboratory :

Tech I (X ₁)	Tech II (X ₂)	Tech III (X ₃)	Tech IV(X ₄)
6	14	10	9
14	9	12	12
10	12	7	8
8	10	15	10
11	14	11	11

Test at the level of significance $\alpha = 0.01$ whether the differences among the 4 samples means can be attributed to chance.

Solution:

H₀: There is no significant difference between the technicians

H ₁ : Significant difference between the technicians										
We shift the origin										
Total	X ₁	X ₂	X ₃	X ₄	TOTAL	X ₁ ²	X ₂ ²	X ₃ ²	X ₄ ²	
	-4	4	0	-1	-1	16	16	0	1	
	4	-1	2	2	7	16	1	4	4	
	0	2	-3	-2	-3	0	4	9	4	
	-2	0	5	0	3	4	0	25	0	
	1	4	1	1	7	1	16	1	1	
	-1	9	5	0	13	37	37	39	10	
N= Total No of Observations = 20										
T=Grand Total = 13										
Correction Factor = $\frac{(\text{Grand total})^2}{\text{Total No of Observations}} = 8.45$										
TSS = $\sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \sum X_4^2 - C.F = 37 + 37 + 39 + 10 - 8.45 = 114.55$										
SSC = $\frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} - C.F = \frac{(-1)^2}{5} + \frac{(9)^2}{5} + \frac{(5)^2}{5} + 0 - 8.45 = 12.95$										
SSE = TSS – SSC = 114.55-12.95= 101.6										
ANOVA Table										
Source of Variation	Sum of Squares	Degree of freedom	Mean Square	F- Ratio						
Between Samples	SSC=12.95	C-1= 4-1=3	MSC = $\frac{SSC}{K - 1} = 4.317$	$F_c = \frac{MSC}{MSE} = 1.471$						
Within Samples	SSE=101.6	N-C=20-4=16	MSE = $\frac{SSE}{N - K} = 6.35$							
Cal F _C = 1.471 & Tab F _C (16,3) = 5.29										
Conclusion : Cal F _C < Tab F _C ⇒ There is no significance difference between the technicians										
2	A completely randomized design exprement with 10 plots and 3 treatments gave the following results.									
Plot No	1	2	3	4	5	6	7	8	9	10
Treatment	A	B	C	A	C	C	A	B	A	B
Yield	5	4	3	7	5	1	3	4	1	7
Analyse the results for treatment effects.										
Solution:										

A	B	C
5	4	3
7	4	5
3	7	1
1		

Null Hypothesis H₀: There is no significant difference in treatments

Alternate Hypothesis H₁ : Significant difference in treatments

Total	X₁	X₂	X₃	TOTAL	X₁²	X₂²	X₃²
	5	4	3	12	25	16	9
	7	4	5	16	49	16	25
	3	7	1	11	9	49	1
	1			1	1		
	16	15	9	40	84	81	35

Step1: N= Total No of Observations = 10

Step 2: T=Grand Total = 40

Step 3: Correction Factor = $\frac{(\text{Grand total})^2}{\text{Total No of Observations}}$ = $\frac{T^2}{N}$ = $\frac{40^2}{10}$ =160

Step 4: TSS= $\sum X_1^2 + \sum X_2^2 + \sum X_3^2 - C.F$ = 84+81+35 –160 = 40

Step 5: SSC= $\frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} - C.F$ = $\frac{(16)^2}{4} + \frac{15^2}{3} + 3 - 160$

SSC = 64+ 75+ 27 –160 = 6

Where N₁= Number of elements in each column

Step 7: SSE=TSS-SSC = 40 – 6 = 34

Step 8: **ANOVA TABLE:**

Source of Variation	Sum of Squares	Degree of freedom	Mean Square	F- Ratio
Between Columns	SSC=6	C-1= 3-1=2	$MSC = \frac{SSC}{C - 1}$ $= \frac{6}{2} = 3$	$F_c = \frac{MSE}{MSC}$ $= \frac{4.86}{3}$ $= 1.62$
Error	SSE=34	N-C=10-3=7	$MSE = \frac{SSE}{N - C}$ $= \frac{34}{87} = 4.86$	

Cal F_C = 1.62

Table value : $F_C(7,2)=19.35$

Conclusion : $\text{Cal } F_C < \text{Tab } F_C$

We accept Null Hypothesis $\Rightarrow$ There is no significance difference in tretments

3 The following table gives the number of articles of a product produced by five different workers using four types of machines.

Workers	Machines			
	P	Q	R	S
A	44	38	47	36
B	46	40	52	43
C	34	36	44	32
D	43	38	46	33
E	38	42	49	39

Test (i) Whether the five workers differ with respect to mean productivity and
(ii) Whether the four machines differ with respect to mean productivity.

Solution: H_0 : There is no significant difference between the Machine types and no significant difference between the Workers

H_1 :Significant difference between the Machine types and no significant difference between the Workers

We shift the origin $X_{ij} = x_{ij} - 46$; $h = 5$; $k = 4$; $N = 20$

	A	B	C	D	Total= T_{i*}	$[T_{i*}^2]/k$	ΣX_{ij}^2
1	-2	-8	1	-10	-19	90.25	169
2	0	-6	6	-3	-3	2.25	81
3	-12	-10	-2	-14	-38	361	444
4	-3	-8	0	-13	-24	144	242
5	-8	-4	3	-7	-16	64	138
Total=T_{*j}	-25	-36	8	-47	-100	661.5	1074
$[T_{*j}^2]/h$	125	259.2	12.8	441.8	838.8		

$T = \text{Grand Total} = -100$

Correction Factor = $\frac{(\text{Grand total})^2}{\text{Total No of Observations}} = \frac{(-100)^2}{20} = 500$

$TSS = \sum_i \sum_j X_{ij}^2 - C.F = 1074 - 500 = 574$

$SSR = \sum_k T_{i*}^2 - C.F = 661.5 - 500 = 161.5$

$$SSC = \frac{\sum T_{*j}^2}{h} - C.F = 838.8 - 500 = 338.8$$

$$SSE = TSS - SSC - SSR = 574 - 161.5 - 338.8 = 73.7$$

ANOVA Table

Source of Variation	Sum of Squares	Degree of freedom	Mean Square	F- Ratio	F _{Tab} Ratio
Between Rows (Workers)	SSR=161.5	h - 1= 4	MSR= 40.375	F _R = 6.574 F _C = 18.388	F _{5%} (4, 12) = 3.26
Between Columns (Machine)	SSC=338.8	k - 1=3	MSC = 112.933		
Residual	SSE = 73.7	(h - 1)(k - 1) = 12	MSE = 6.1417		F _{5%} (3, 12) = 3.59
Total	1074				

Conclusion : Cal F_C< Tab F_C and Cal F_R< Tab F_R ⇒ There is no significant difference between the Machine types and no significant difference between the Workers

4 A Company appointments four salesmen A, B, C and D and observes their sales in 3 seasons: summer, winter and monsoon. The figures (in lakhs of Rs.) are given in the following table:

Season	Salesman			
	A	B	C	D
Summer	36	36	21	35
Winter	28	29	31	32
Monsoon	26	28	29	29

i) Do the salesmen significantly differ in performance?
ii) Is there significant difference between the seasons?
Solution:
Null Hypothesis H_0 : There is no significant difference between the sales in the 3 seasons and also between the sales of the 4 salesmen.
Alternate Hypothesis H_1 : There is a significant difference between the sales in the 3 seasons and also between the sales of the 4 salesmen.
Test statistic:
To simplify calculations we deduct 30 from each value

Seasons		A X ₁	B X ₂	C X ₃	D X ₄	Seasons Total	X ₁ ²	X ₂ ²	X ₃ ²	X ₄ ²
Y ₁	Summer	6	6	-9	5	8	36	36	81	25
Y ₂	Winter	-2	-1	1	2	0	4	1	1	4
Y ₃	Monson	-4	-2	-1	-1	-8	16	4	1	1
Total		0	3	-9	6	0	56	41	83	30

Step1: N= Total No of Observations = 12

Step 2: T=Grand Total = 0

Step 3: Correction Factor = $\frac{(\text{Grand total})^2}{\text{Total No of Observations}} = \frac{T^2}{N} = \frac{0^2}{12} = 0$

Step 4: TSS= $\sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \sum X_4^2 - C.F = 56 + 41 + 83 + 30 - 0 = 210$

Step 5:

SSC= $\frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} + \frac{(\sum X_4)^2}{N_1} - C.F = \frac{0^2}{3} + \frac{3^2}{3} + \frac{(-9)^2}{3} + \frac{6^2}{3} - 0$

SSC= 0 + 3 + 27 + 12 - 0 = 42

Where N₁= Number of elements in each column

Step 6:

SSR= $\frac{(\sum Y_1)^2}{N_2} + \frac{(\sum Y_2)^2}{N_2} + \frac{(\sum Y_3)^2}{N_2} - C.F = \frac{8^2}{4} + \frac{0^2}{4} + \frac{(-8)^2}{4} + \frac{6^2}{4} - 0 = 16 + 0 + 16 - 0 = 32$

Where N₂= Number of elements in each row

Step 7: SSE=TSS-SSC-SSR = 210 - 42 - 32

Step 8: ANOVA TABLE:

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Sum of Squares	variance	F – ratio	
Between Columns (Salesmen)	SSC=42	c-1=4-1=3	$MSC = \frac{SSC}{c-1}$ $= \frac{42}{3} = 14$	$MSC = \frac{MSE}{MSC}$ $= \frac{22.67}{14}$ $= 1.619$	$F_C(6,3) = 8.94$	
Between rows (Seasons)	SSR =32	r-1=3-1=2	$MSR = \frac{SSR}{r-1}$ $= \frac{32}{2} = 16$	$MSR = \frac{MSE}{MSR}$ $= \frac{22.67}{16}$ $= 1.417$	$F_R(6,2) = 8.94$	

	Error	SSE=136	N-c-r +1=6	$MSE = \frac{SSE}{N - c - r + 1}$ $= \frac{136}{6} = 22.67$			
	Total	210	11				

Table Value of F = $F_C(\text{Error,d.f}) = F_C(6,3) = 8.94$, $F_R(\text{Error,d.f}) = 8.94$ with 5% level of significance

Conclusion:

1) Cal $F_R < \text{Table } F_{R,0.05}(6,3)$

Hence we accept the H_0 and we conclude that there is no significant difference between sales in the three seasons.

2) Cal $F_R < \text{Table } F_{R,0.05}(6,2)$.

Hence we accept the H_0 and we conclude that there is no significant difference between in the sales of 4 salesmen.

5

Analyze 2² factorial experiments for the following table.

Treatment	Replications			
	I	II	III	IV
(1)	12	12.3	11.8	11.6
a	12.8	12.6	13.7	14
b	11.5	11.9	12.6	11.8
ab	14.2	14.5	14.4	15

SOLUTION:

Null hypothesis: All the mean effects are equal.

Let A and B be the two factors.

Let n=number of replications=4

Subtract 12 from each

Treatment	Replications			
	I	II	III	IV
(1)	0	0.3	-0.2	-0.4
a	0.8	0.6	1.7	2
b	-0.5	-0.1	0.6	-0.2
ab	2.2	2.5	2.4	3

Let us find SS for the table

Treatment	Replications				Row Total R_i	R_i^2
	I	II	III	IV		
(1)	0	0.3	-0.2	-0.4	-0.3	0.09
a	0.8	0.6	1.7	2	5.1	26.01
b	-0.5	-0.1	0.6	-0.2	-0.2	0.04
ab	2.2	2.5	2.4	3	10.1	102.01
Column Total C_j	2.5	3.3	4.5	4.4	T=14.7	
C_j^2	6.25	10.89	20.25	19.36		

T=14.7

Correction factor= $\frac{T^2}{N}$ =13.5

TSS=21.19

SSC=0.688

SSR=18.54

SSE=1.962

Source of Variation	Sum of Squares	Degree of freedom	Mean Square	F- Ratio	F _{Tab} Ratio
b	S_B =1.63	1	MSB=1.63	$F_B = 7.409$	10.56
a	S_A =15.41	1	MSA=15.41	$F_A = 70.04$	10.56

ab	$S_{AB}=1.50$	1	MSAB=1.50	$F_{AB} = 6.81$	10.56
Error	SSE=1.962	N-C-r+1=9	SSE=1.962		

Cal(F_A)=70.04 $\Rightarrow H_0$ is rejected at 1% level

Cal(F_B)=7.409 $\Rightarrow H_0$ is accepted at 1% level

Cal(F_{AB})=10.56 $\Rightarrow H_0$ is accepted at 1% level

6 . **Analyse the variance in the following latin square of yields (in kgs) of paddy where A, B, C, D denote the different methods of cultivation.**

D 122 A 121 C 123 B 122
B 124 C 123 A 122 D 125
A 120 B 119 D 120 C 121
C 122 D 123 B 121 A 122

Examine whether the different methods of cultivation have given significantly different yields.

Solution:

We shift the origin $X_{ij} = x_{ij} - 100$; n = 4; N = 16

	I	II	III	IV	Total= T_i^*	$[T_i^{*2}]/n$	ΣX_{ij}^2
A	2	1	3	2	8	16	18
B	4	3	2	5	14	49	54
C	0	-1	0	1	0	0	2
D	2	3	1	2	8	16	18
Total=T_{*j}	8	6	6	10	30	81	92
$[T_{*j}^2]/n$	16	9	9	25	59		
ΣX_i^{*2}	24	20	14	34	92		

	Letters				Total=T _i *	[T _i *] ² /n
P	1	2	0	2	5	6.25
Q	2	4	-1	1	6	9
R	3	3	1	2	9	20.25
S	2	5	0	3	10	25
Total					30	60.5

T=Grand Total = 30 ;Correction Factor = $\frac{(Grand\ total)^2}{Total\ No\ of\ Observations} = \frac{(30)^2}{16}$

$TSS = \sum_i \sum_j X_{ij}^2 - C.F = 92 - \frac{(30)^2}{16} = 35.75$

$SSR = \frac{\sum T_i^2}{n} - C.F = 81 - \frac{(30)^2}{16} = 24.75$

$SSC = \frac{\sum T_j^2}{n} - C.F = 59 - \frac{(30)^2}{16} = 2.75$

$SSL = \frac{\sum T_{r*}^2}{n} - C.F = 60.5 - \frac{(30)^2}{16} = 4.25$

$SSE = TSS - SSC - SSR - SSL = 35.75 - 24.75 - 2.75 - 4.25 = 4$

ANOVA Table

Source of Variation	Sum of Squares	Degree of freedom	Mean Square	F- Ratio	F _{Tab} Ratio (5% level)
Between Rows	SSR=24.75	n - 1= 3	MSR=8.25	F _R = 12.31 F _C = 1.37 F _L = 2.12	F _R (3, 6)=4.76 F _c (3, 6)=4.76
Between Columns	SSC=2.75	n - 1= 3	MSC = 0.92		
Between Letters	SSL = 4.25	n - 1= 3	MSL = 1.42		
Residual	SSE= 4	(n – 1)(n – 2) = 6	MSE = 0.67		

						$F_L(3, 6)=4$.76
Total	35.75					

Conclusion :
Cal $F_C < \text{Tab } F_C$, Cal $F_L < \text{Tab } F_L$ and Cal $F_R > \text{Tab } F_R \Rightarrow$ There is significant difference between the **rows** , no significant difference between the letters and no significant difference between the columns

7

A variable trial was conducted on wheat with 4 varieties in a Latin Square Design. The plan of the experiment and the per plot yield are given below :

C	25	B	23	A	20	D	20
A	19	D	19	C	21	B	18
B	19	A	14	D	17	C	20
D	17	C	20	B	21	A	15

Analyse data and interpret the result.
 H_0 : Four varieties are similar
 H_1 : Four varieties are not similar
Let us take 20 as origin for simplifying the calculation

Variety	X_1	X_2	X_3	X_4	TOTAL	X_1^2	X_2^2	X_3^2	X_4^2
Y_1	5	3	0	0	8	25	9	0	0
Y_2	-1	-1	1	-2	-3	1	1	1	4
Y_3	-1	-6	-3	0	-10	1	36	9	0
Y_4	-3	0	1	-5	-7	1	0	1	25
	0	-4	-1	-7	-12	9	46	11	29

N=Total No of Observations = 16

T=Grand Total = -12

Correction Factor = $\frac{(\text{Grand total})^2}{\text{Total No of Observations}} = 9$

$$TSS = \sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \sum X_4^2 - C.F = 36 + 46 + 11 + 29 - 9 = 113$$

$$SSC = \frac{(\sum X_1)_2}{N_1} + \frac{(\sum X_2)_2}{N_1} + \frac{(\sum X_3)_2}{N_1} - C.F = \frac{(6)^2}{4} + \frac{(10)^2}{4} + \frac{(6)^2}{4} + \frac{(10)^2}{4} - 9 = 4$$

$$SSR = \frac{(\sum Y_1)^2}{N_1} + \frac{(\sum Y_2)^2}{N_2} + \frac{(\sum Y_3)^2}{N_2} + \frac{(\sum Y_4)^2}{N_2} - C.F = \frac{(8)^2}{4} + \frac{(-3)^2}{4} + \frac{(-10)^2}{4} + \frac{(-7)^2}{4} - 9 = 46.5$$

To find SSK

Treatment	1	2	3	4	Total
A	0	-1	-6	-5	-12
B	3	-2	-1	1	1
C	5	1	0	0	6
D	0	-1	-3	-3	-7

$$SSK = \frac{(\sum Y_1)^2}{K_1} + \frac{(\sum Y_2)^2}{K_2} + \frac{(\sum Y_3)^2}{K_3} + \frac{(\sum Y_4)^2}{K_4} - C.F$$

$$= 48.5$$

$$SSE = TSS - SSC - SSR - SSK = 113 - 7.5 - 46.5 - 48.5 = 10.5$$

ANOVA Table

Source of Variation	Sum of Squares	Degree of freedom	Mean Square	F- Ratio
Column Treatment	SSC=7.5	n-1=3	$MSC = \frac{SSC}{n - 1}$ $= 2.5$	$F_c = \frac{MSC}{MSE} = 1.43$
Row Treatments	SSR=46.5	n-1=3	$MSR = \frac{SSR}{n - 1}$ $= 15.5$	$F_r = \frac{MSE}{MSR} = 8.86$
Between Treatments	SST=48.5	n-1=3	$MSK = \frac{SSK}{n - 1}$ $= 16.17$	$F_k = \frac{MSK}{MSE} = 9.24$
Error (or) Residual	SSE=10.5	(n-1) (n-2)=6	MSE $= \frac{SSE}{(n - 1)(n - 2)}$ $= 1.75$	

Table value F(3,6) degrees of freedom 8.94

There is significant difference between treatments